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Pivotal bone anchor assembly with independent provisional locking

Abstract

A pivotal bone anchor assembly includes a receiver having a cavity with a bottom opening and a first channel for receiving a rod, and a shank having a head positionable into the cavity with the shank extending downward through the bottom opening. The assembly also includes an insert positionable into the central bore above the shank head and having a second channel alignable with the first channel, opposite extensions extending outwardly in the direction of the channels to block rotation between the insert and the receiver, and upward-facing surfaces located laterally outward from and on each side of the second open channel. The upward-facing surfaces are configured for releasable compressive engagement by a compressing tool after the elongate rod is received within the channels and prior to a final locking of the assembly so as to provisionally lock the shank from pivoting with respect to the receiver.

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us-provisional-application US 61456649 20101110
us-provisional-application US 61403915 20100923
us-provisional-application US 61403696 20100920
us-provisional-application US 61402959 20100908
us-provisional-application US 61400504 20100729
us-provisional-application US 61398807 20100701
us-provisional-application US 61396390 20100526
us-provisional-application US 61395752 20100517
us-provisional-application US 61395564 20100514
us-provisional-application US 61343737 20100503
us-provisional-application US 61336911 20100128
us-provisional-application US 61278240 20091005
us-provisional-application US 61270754 20090713
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5669911	12/1996	Errico et al.	N/A	N/A
5672176	12/1996	Biedermann et al.	N/A	N/A
5690630	12/1996	Errico et al.	N/A	N/A
5733286	12/1997	Errico et al.	N/A	N/A
5782833	12/1997	Haider	N/A	N/A

5797911	12/1997	Sherman et al.	N/A	N/A
5885286	12/1998	Sherman et al.	N/A	N/A
6010503	12/1999	Richelsoph et al.	N/A	N/A
6063090	12/1999	Schläpfer	N/A	N/A
6074391	12/1999	Metz-Stavenhagen et al.	N/A	N/A
6090111	12/1999	Nichols	N/A	N/A
6187005	12/2000	Brace et al.	N/A	N/A
6254602	12/2000	Justis	N/A	N/A
6280442	12/2000	Barker et al.	N/A	N/A
6355040	12/2001	Richelsoph et al.	N/A	N/A
6471705	12/2001	Biedermann et al.	N/A	N/A
6485491	12/2001	Farris et al.	N/A	N/A
6485494	12/2001	Haider	N/A	N/A
6530929	12/2002	Justis et al.	N/A	N/A
6540748	12/2002	Lombardo	N/A	N/A
6554834	12/2002	Crozet et al.	N/A	N/A
6565565	12/2002	Yuan et al.	N/A	N/A
6626908	12/2002	Cooper et al.	N/A	N/A
6648888	12/2002	Shluzas	N/A	N/A
6723100	12/2003	Biedermann et al.	N/A	N/A
6740086	12/2003	Richelsoph	N/A	N/A
6835093	12/2003	Biedermann et al.	N/A	N/A
6835196	12/2003	Biedermann et al.	N/A	N/A
6837889	12/2004	Shluzas	N/A	N/A
6905500	12/2004	Jeon et al.	N/A	N/A
6945975	12/2004	Dalton	N/A	N/A
7066937	12/2005	Shluzas	N/A	N/A
7087057	12/2005	Konieczynski et al.	N/A	N/A
7141051	12/2005	Janowski et al.	N/A	N/A
7144396	12/2005	Shluzas	N/A	N/A
7163539	12/2006	Abdelgany et al.	N/A	N/A
7179261	12/2006	Sicvol et al.	N/A	N/A
7264621	12/2006	Coates et al.	N/A	N/A
7306606	12/2006	Sasing	N/A	N/A
7311712	12/2006	Dalton	N/A	N/A
7377923	12/2007	Purcell et al.	N/A	N/A
7445627	12/2007	Hawkes et al.	N/A	N/A
7479156	12/2008	Lourdel et al.	N/A	N/A
7604655	12/2008	Warnick	N/A	N/A
7618444	12/2008	Shluzas	N/A	N/A
7662172	12/2009	Warnick	N/A	N/A
7686834	12/2009	Saint Martin	N/A	N/A
7686835	12/2009	Warnick	N/A	N/A
7695497	12/2009	Cordaro et al.	N/A	N/A
7699876	12/2009	Barry et al.	N/A	N/A
7722654	12/2009	Taylor et al.	N/A	N/A
7766915	12/2009	Jackson	N/A	N/A
7766945	12/2009	Nilsson et al.	N/A	N/A
7776067	12/2009	Jackson	N/A	N/A

7789896	12/2009	Jackson	N/A	N/A
7789900	12/2009	Levy et al.	N/A	N/A
7811310	12/2009	Baker et al.	N/A	N/A
7833251	12/2009	Ahlgren et al.	N/A	N/A
7842073	12/2009	Richelsoph et al.	N/A	N/A
7857834	12/2009	Boschert	N/A	N/A
7901436	12/2010	Baccelli	N/A	N/A
7905907	12/2010	Spitler et al.	N/A	N/A
7909830	12/2010	Frigg et al.	N/A	N/A
7914536	12/2010	MacDonald et al.	N/A	N/A
7935135	12/2010	Mujwid	N/A	N/A
7942909	12/2010	Hammill, Sr. et al.	N/A	N/A
7947065	12/2010	Hammill, Sr. et al.	N/A	N/A
7951172	12/2010	Chao et al.	N/A	N/A
7951173	12/2010	Hammill, Sr. et al.	N/A	N/A
7955359	12/2010	Matthis et al.	N/A	N/A
7967850	12/2010	Jackson	N/A	N/A
7972364	12/2010	Biedermann et al.	N/A	N/A
7988694	12/2010	Barrus et al.	N/A	N/A
8021397	12/2010	Farris et al.	N/A	N/A
8048112	12/2010	Suzuki et al.	N/A	N/A
8066744	12/2010	Justis et al.	N/A	N/A
8075603	12/2010	Hammill, Sr. et al.	N/A	N/A
8083776	12/2010	Alvarez	N/A	N/A
8097025	12/2011	Hawkes et al.	N/A	N/A
8162985	12/2011	Kim	N/A	N/A
8197517	12/2011	Lab et al.	N/A	N/A
8197518	12/2011	Hammill, Sr. et al.	N/A	N/A
8221472	12/2011	Peterson et al.	N/A	N/A
8308782	12/2011	Jackson	N/A	N/A
8328817	12/2011	Strauss	N/A	N/A
8353932	12/2012	Jackson	N/A	N/A
8361123	12/2012	Fanger et al.	N/A	N/A
8382805	12/2012	Wang et al.	N/A	N/A
8398683	12/2012	Berrevoets et al.	N/A	N/A
8439922	12/2012	Arnold et al.	N/A	N/A
8444681	12/2012	Jackson et al.	N/A	N/A
8449578	12/2012	Keiser et al.	N/A	N/A
8556938	12/2012	Jackson et al.	N/A	N/A
8562652	12/2012	Biedermann et al.	N/A	N/A
8628558	12/2013	Harvey et al.	N/A	N/A
8663290	12/2013	Doubler et al.	N/A	N/A
8663291	12/2013	Doubler et al.	N/A	N/A
8663298	12/2013	Keyer et al.	N/A	N/A
8696712	12/2013	Biedermann et al.	N/A	N/A
8876869	12/2013	Schafer et al.	N/A	N/A
8882817	12/2013	Jones et al.	N/A	N/A
8888827	12/2013	Harper et al.	N/A	N/A
8926671	12/2014	Biedermann et al.	N/A	N/A
8986349	12/2014	German et al.	N/A	N/A

8998959	12/2014	Jackson et al.	N/A	N/A
9044272	12/2014	Shaffrey et al.	N/A	N/A
9078705	12/2014	Matthis	N/A	A61B 17/7002
9119674	12/2014	Matthis et al.	N/A	N/A
9155567	12/2014	Auerbach et al.	N/A	N/A
9168069	12/2014	Jackson et al.	N/A	N/A
9198695	12/2014	Shluzas et al.	N/A	N/A
9254150	12/2015	Biedermann et al.	N/A	N/A
9259247	12/2015	Chandanson et al.	N/A	N/A
9358047	12/2015	Mishra et al.	N/A	N/A
9445847	12/2015	Biedermann et al.	N/A	N/A
9480517	12/2015	Jackson et al.	N/A	N/A
9486246	12/2015	Biedermann et al.	N/A	N/A
9566092	12/2016	Jackson et al.	N/A	N/A
9572600	12/2016	Biedermann et al.	N/A	N/A
9603627	12/2016	Krüger	N/A	N/A
9655652	12/2016	Biedermann et al.	N/A	N/A
9763702	12/2016	Schlaepfer et al.	N/A	N/A
9907574	12/2017	Jackson et al.	N/A	N/A
9918745	12/2017	Jackson et al.	N/A	N/A
9980753	12/2017	Jackson et al.	N/A	N/A
10052136	12/2017	Nelson	N/A	N/A
10154859	12/2017	Keyer et al.	N/A	N/A
10172647	12/2018	Elsbury	N/A	N/A
10219837	12/2018	Jackson et al.	N/A	N/A
10363070	12/2018	Jackson et al.	N/A	N/A
10478225	12/2018	Jackson	N/A	A61B 17/702
10792074	12/2019	Jackson	N/A	N/A
10869694	12/2019	Jackson et al.	N/A	N/A
10898233	12/2020	Jackson et al.	N/A	N/A
10987137	12/2020	Jackson et al.	N/A	N/A
11147592	12/2020	Jackson	N/A	A61B 17/7032
11234745	12/2021	Jackson	N/A	N/A
11464548	12/2021	Jackson et al.	N/A	N/A
11484346	12/2021	Jackson et al.	N/A	N/A
11571244	12/2022	Loftis et al.	N/A	N/A
12016594	12/2023	Jackson et al.	N/A	N/A
2003/0149431	12/2002	Varieur	N/A	N/A
2004/0102781	12/2003	Jeon	N/A	N/A
2004/0122425	12/2003	Suzuki	606/267	A61B 17/7062
2004/0186473	12/2003	Cournoyer et al.	N/A	N/A
2004/0260283	12/2003	Wu et al.	N/A	N/A
2005/0203516	12/2004	Biedermann et al.	N/A	N/A
2006/0058788	12/2005	Hammer et al.	N/A	N/A
2006/0161152	12/2005	Ensign et al.	N/A	N/A
2006/0161153	12/2005	Hawkes et al.	N/A	N/A
2006/0217716	12/2005	Baker et al.	N/A	N/A
2006/0276789	12/2005	Jackson	606/301	A61B 17/70
2007/0118123	12/2006	Strausbaugh et al.	N/A	N/A
2007/0270813	12/2006	Garamszegi	N/A	N/A

2010/0152787	12/2009	Walsh	606/305	A61B 17/7037
2010/0234902	12/2009	Biedermann	606/305	A61B 17/7037
2010/0331887	12/2009	Jackson et al.	N/A	N/A
2011/0196430	12/2010	Walsh	606/305	A61B 17/7037
2011/0270321	12/2010	Prevost et al.	N/A	N/A
2012/0310284	12/2011	Gerchow	N/A	N/A
2013/0103098	12/2012	Jackson et al.	N/A	N/A
2013/0218213	12/2012	Lemoine	N/A	N/A
2014/0025119	12/2013	Biedermann et al.	N/A	N/A
2021/0236174	12/2020	Jackson et al.	N/A	N/A
2022/0387081	12/2021	Jackson et al.	N/A	N/A
2023/0414256	12/2022	Jackson et al.	N/A	N/A
2024/0032972	12/2023	Jackson et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
1857064	12/2006	EP	N/A
WO 2009/055747	12/2008	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/504,135 filed Oct. 18, 2021, which is a continuation of U.S. patent application Ser. No. 17/075,955, now U.S. Pat. No. 11,147,592 filed Oct. 21, 2020, which is a continuation of U.S. patent application Ser. No. 16/593,086, filed Oct. 4, 2019, now U.S. Pat. No. 10,813,672, which is a continuation of U.S. patent application Ser. No. 16/591,457, filed Oct. 2, 2019, now U.S. Pat. No. 10,856,911, which is a continuation of U.S. patent application Ser. No. 16/514,798, filed Jul. 17, 2019, now U.S. Pat. No. 10,813,671, which is a continuation of U.S. patent application Ser. No. 16/393,544, filed Apr. 24, 2019, now U.S. Pat. No. 10,945,768, which is a continuation of U.S. patent application Ser. No. 15/969,502, filed May 2, 2018, now U.S. Pat. No. 10,278,738, which is a continuation of U.S. patent application Ser. No. 13/374,439, filed Dec. 29, 2011, now U.S. Pat. No. 9,980,753, which claims the benefit of U.S. Provisional Patent Application Ser. No. 61/460,267, filed Dec. 29, 2010, and U.S. Provisional Patent Application Ser. No. 61/463,037, filed Feb. 11, 2011, each of which is incorporated by reference in its entirety herein. (2) Application Ser. No. 13/374,439 is also a continuation-in-part of U.S. patent application Ser. No. 13/373,289, filed Nov. 9, 2011, now U.S. Pat. No. 9,907,574, which claims the benefit of U.S. Provisional Patent Application Ser. No. 61/456,649, filed Nov. 10, 2010, and U.S. Provisional Patent Application Ser. No. 61/460,234, filed Dec. 29, 2010, each of which is incorporated by reference in its entirety herein. (3) Application Ser. No. 13/374,439 is also a continuation-in-part of U.S. patent application Ser. No. 12/924,802 filed Oct. 5, 2010, now U.S. Pat. No. 8,556,938, which claims the benefit of the following U.S. Provisional Patent Application Ser. Nos. 61/278,240, filed Oct. 5, 2009; 61/336,911, filed Jan. 28, 2010; 61/343,737 filed May 3, 2010; 61/395,564, filed May 14, 2010; 61/395,752, filed May 17, 2010; 61/396,390, filed May 26, 2010; 61/398,807, filed Jul. 1, 2010; 61/400,504, filed Jul. 29, 2010; 61/402,959, filed Sep. 8, 2010; 61/403,696, filed Sep. 20, 2010; and 61/403,915, filed Sep. 23, 2010, each of which is incorporated by reference in its entirety herein. (4) Application Ser. No. 13/374,439 is also a continuation-in-part of U.S. patent application Ser. No. 12/802,849 filed Jun. 15, 2010, now abandoned, which claims the benefit of the following U.S. Provisional Patent Application Ser. Nos. 61/396,390 filed May 26, 2010; 61/395,752 filed May 17, 2010; 61/395,564 filed May 14, 2010; 61/336,911 filed Jan. 28, 2010; 61/270,754 filed Jul. 13, 2009; and 61/268,708 filed Jun. 15, 2009, each of which is incorporated by reference in its entirety herein.

BACKGROUND OF THE INVENTION

(1) The present invention is directed to polyaxial bone screw shanks with heads for use in bone surgery, more specifically to spinal surgery and particularly to such screws with receiver member assemblies including compression or pressure inserts and expansion-only split retainers to snap over, capture and retain the bone screw shank head in the receiver member assembly and later fix the bone screw shank with respect to the receiver assembly.

(2) Bone screws are utilized in many types of spinal Surgery in order to secure various implants to vertebrae along the spinal column for the purpose of stabilizing and/or adjusting spinal alignment. Although both closed-ended and open-ended bone screws are known, open-ended screws are particularly well suited for connections to rods and connector arms, because such rods or arms do not need to be passed through a closed bore, but rather can be laid or urged into an open channel within a receiver or head of such a screw. Generally, the screws must be inserted into the bone as an integral unit along with the head, or as a preassembled unit in the form of a shank and pivotal receiver, such as a polyaxial bone screw assembly.

(3) Typical open-ended bone screws include a threaded shank with a pair of parallel projecting branches or arms which form a yoke with a U-shaped slot or channel to receive a rod. Hooks and other types of connectors, as are used in spinal fixation techniques, may also include similar open ends for receiving rods or portions of other fixation and stabilization structure.

(4) A common approach for providing vertebral column support is to implant bone screws into certain bones which then in turn support a longitudinal structure such as a rod, or are supported by such a rod. Bone screws of this type may have a fixed head or receiver relative to a shank thereof, or may be of a polyaxial screw nature. In the fixed bone screws, the rod receiver head cannot be moved relative to the shank and the rod must be favorably positioned in order for it to be placed within the receiver head. This is sometimes very difficult or impossible to do. Therefore, polyaxial bone screws are commonly preferred. Open-ended polyaxial bone screws typically allow for a loose or floppy rotation of the head or receiver about the shank until a desired rotational position of the receiver is achieved by fixing such position relative to the shank during a final stage of a medical procedure when a rod or other longitudinal connecting member is inserted into the receiver, followed by a locking screw or other closure. This floppy feature can be, in some cases, undesirable and make the procedure more difficult. Also, it is often desirable to insert the bone screw shank separate from the receiver or head due to its bulk which can get in the way of what the surgeon needs to do. Such screws that allow for this capability are sometimes referred to as modular polyaxial screws.

(5) With specific reference to modular snap-on or pop-on polyaxial pedicle screw systems having shank receiver assemblies, the prior art has shown and taught the concept of the receiver and certain retainer parts forming an assembly wherein a contractile locking engagement between the parts is created to fix the shank head with respect to the receiver and retainer. The receiver and shank head retainer assemblies in the prior art have included a contractile retainer ring and/or a lower pressure insert with an expansion and contraction collet-type of structure having contractile locking engagement for the shank head due to direct contact between the retainer and/or the collet structure with the receiver resulting in contraction of the retainer ring and/or the collet-type structure of the insert against the shank head.

(6) The prior art for modular polyaxial screw assemblies has also shown and taught that the contact surfaces on the outside of the collect and/or retainer and the inside of the receiver can be tapered, conical, radiused, spherical, curvate, multi-curvate, rounded, as well as other configurations to create a contractile type of locking engagement for the shank head with respect to the receiver.
(7) In addition, the prior art for modular polyaxial screw assemblies has shown and taught that the shank head can both enter and escape from a collet-like structure on the insert or from the retainer when the insert or retainer is in the up position and within an expansion recess or chamber of the receiver. This is the case unless the insert and/or the retainer are blocked from being able to be pushed back up into receiver bore or cavity.

SUMMARY OF THE INVENTION

(8) The present invention differentiates from the prior art by not allowing the receiver to be removed from the shank head once the parts are snapped-on and connected. This is true even if the retainer can go back up into the expansion chamber. This approach or design has been found to be more secure and to provide more resistance to pull-out forces compared to the prior art for modular polyaxial screw designs. Collect-like structures extending downwardly from lower pressure inserts, when used in modular polyaxial screw designs, as shown in the prior art, have been found to be somewhat weak with respect to pull-out forces encountered during some spinal reduction procedures. The present invention is designed to solve these problems.
(9) The present invention also differentiates from all of the prior art by providing a split retainer ring and a cooperating insert with a collet-like lower super structure portion, wherein the super structure does not participate at all in the locking engagement for the shank head with respect to the receiver. The expansion-only split or open retainer ring in the present invention is positioned entirely below the shank head hemisphere in the receiver and can be a stronger, more substantial structure to resist larger pull out forces on the assembly. The retainer ring base can also be better supported on a generally horizontal loading surface near the lower opening in the bottom of the receiver. This design has been found to be stronger and more secure when compared to that of the prior art which uses some type of contractile locking engagement between the parts, as described above; and, again, once assembled it cannot be disassembled.
(10) Thus, a polyaxial bone screw assembly according to the invention includes a shank having an integral upper portion or head and a body for fixation to a bone; a separate receiver defining an upper open channel, a central bore, a lower cavity and a lower opening; one of (a) a top drop and rotate friction fit lower compression insert having super structure providing temporary friction fit with the shank head and (b) a top drop, non-rotatable friction fit insert having a seating surface extending between front and rear faces of arms of the receiver and super structure providing temporary friction fit with the shank head; and a resilient expansion-only split retainer for capturing the shank head in the receiver lower cavity, the shank head being frictionally engaged with, but still movable in a non-floppy manner with respect to the friction fit insert and the receiver prior to locking of the shank into a desired configuration. In some embodiments the insert completely extends between receiver arms that define the receiver channel and thus remains aligned with and cannot rotate with respect to the receiver at any stage of assembly. In other embodiments, the top drop and rotate insert is fixed with respect to the receiver by pressing or crimping portions of the receiver against the insert. The insert operatively engages the shank head and is spaced from the retainer by the shank head that is snapped into insert super structure illustrated as resilient panels. The shank is finally locked into a fixed position relative to the receiver by frictional engagement between a portion of the insert located above the super structure and the split retainer, as described previously, due to a downward force placed on the compression insert by a closure top pressing on a rod, or other longitudinal connecting member, captured within the receiver bore and channel. In the illustrated embodiments, retainers and inserts are downloaded into the receiver, but uploaded retainer embodiments are also foreseen. The shank head can be positioned into the receiver lower cavity at the lower opening thereof prior to or after insertion of the shank into bone. Some

compression inserts include a lock and release feature for independent locking of the polyaxial mechanism so the screw can be used like a fixed monoaxial screw. The shank can be cannulated for minimally invasive surgery applications. The receiver is devoid of any type of spring tabs or collet-like structures. The pressure insert and/or the retainer are both devoid of any type of receiver-retainer contractile locking engagements with respect to the shank head. In some embodiments, the insert can also have resilient outwardly and upwardly extending arms which are deployed into openings in the receiver cavity so that the retainer and captured shank head are stabilized and retained in the region of the receiver locking chamber once they enter into such lower portion of the receiver cavity. In this way, the shank head and retainer cannot go back up into the receiver cavity.

(11) Again, a pre-assembled receiver, friction fit compression insert and split retainer may be “pushed-on”, “snapped-on” or “popped-on” to the shank head prior to or after implantation of the shank into a vertebra. Such a “snapping on” procedure includes the steps of uploading the shank head into the receiver lower opening, the shank head pressing against the base of the split retainer ring and expanding the resilient retainer out into an expansion portion or chamber of the receiver cavity followed by an elastic return of the retainer back to an original or near nominal shape thereof after the hemisphere of the shank head or upper portion passes through the ring-like retainer. The shank head also enters into the friction fit lower portion of the insert, the panels of the friction fit portion of the insert snapping onto the shank head as the retainer returns to a neutral or close to neutral orientation, providing a non-floppy connection between the insert and the shank head. The friction fit between the shank head and the insert is temporary and not part of the final locking mechanism. In the illustrated embodiment, when the shank is ultimately locked between the compression insert and the lower portion of the retainer, the friction fit collet-like panels of the insert are no longer in a tight friction fit engagement with the shank head and they are not in contact with the receiver. The final fixation occurs as a result of a locking expansion-type of contact between the shank head and the split retainer and an expansion-type of non-tapered locking engagement between the retainer ring and the locking chamber in the lower portion of the receiver cavity. The retainer can expand more in the upper portion or expansion chamber of the receiver cavity to allow the shank head to pass through, but has restricted expansion to retain the shank head when the retainer is against the locking chamber surfaces in the lower portion of the receiver cavity and the shank head is forced down against the retainer ring during final locking. In some embodiments, when the polyaxial mechanism is locked, portions of the insert are forced or wedged against opposing surfaces of the receiver resulting in an interference, non-contractile locking engagement, allowing for adjustment or removal of the rod or other connecting member without loss of a desired angular relationship between the shank and the receiver. This independent, non-contractile locking feature allows the polyaxial screw to function like a fixed monoaxial screw.

(12) The lower pressure insert may also be configured to be independently locked by a tool or instrument, thereby allowing the pop-on polyaxial screw to be distracted, compressed and/or rotated along and around the rod to provide for improved spinal correction techniques. Such a tool engages the pop-on receiver from the sides and then engages the insert to force the insert down into a locked position within the receiver. With the tool still in place and the correction maintained, the rod is then locked within the receiver channel by a closure top followed by removal of the tool. This process may involve multiple screws all being manipulated simultaneously with multiple tools to achieve the desired correction.

(13) Objects of the invention further include providing apparatus and methods that are easy to use and especially adapted for the intended use thereof and wherein the tools are comparatively inexpensive to produce. Other objects and advantages of this invention will become apparent from the following description taken in conjunction with the accompanying drawings wherein are set forth, by way of illustration and example, certain embodiments of this invention.

(14) The drawings constitute a part of this specification and include exemplary embodiments of the present invention and illustrate various objects and features thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is an exploded front elevational view of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, an open expansion-only retainer and a friction fit lower compression insert, further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

(2) FIG. **2** is an enlarged top plan view of the shank of FIG. **1**.

(3) FIG. **3** is reduced cross-sectional view taken along the line **3-3** of FIG. **2**.

(4) FIG. **4** is an enlarged side elevational view of the receiver of FIG. **1**.

(5) FIG. **5** is an enlarged perspective view of the receiver of FIG. **4**.

(6) FIG. **6** is an enlarged top plan view of the receiver of FIG. **4**.

(7) FIG. **7** is an enlarged bottom plan view of the receiver of FIG. **4**.

(8) FIG. **8** is an enlarged cross-sectional view taken along the line **8-8** of FIG. **6**.

(9) FIG. **9** is an enlarged cross-sectional view taken along the line **9-9** of FIG. **6**.

(10) FIG. **10** is an enlarged perspective view of the retainer of FIG. **1**.

(11) FIG. **11** is a reduced top plan view of the retainer of FIG. **10**.

(12) FIG. **12** is a bottom plan view of the retainer of FIG. **10**.

(13) FIG. **13** is an enlarged cross-sectional view taken along the line **13-13** of FIG. **11**.

(14) FIG. **14** is an enlarged perspective view of the insert of FIG. **1**.

(15) FIG. **15** is another enlarged perspective view of the insert of FIG. **1**.

(16) FIG. **16** is a front elevational view of the insert of FIG. **14**.

(17) FIG. **17** is a reduced top plan view of the insert of FIG. **14**.

(18) FIG. **18** is a bottom plan view of the insert of FIG. **14**.

(19) FIG. **19** is an enlarged cross-sectional view taken along the line **19-19** of FIG. **17**.

(20) FIG. **20** is an enlarged cross-sectional view taken along the line **20-20** of FIG. **17**.

(21) FIG. **21** is an enlarged front elevational view of the retainer and receiver of FIG. **1** with portions of the receiver broken away to show the detail thereof, the retainer being shown downloaded into the receiver (in phantom) to an inserted stage of assembly.

(22) FIG. **22** is a front elevational view of the retainer and receiver with portions broken away, similar to what is shown in FIG. **21**, showing the insert of FIG. **1**, also in front elevational view, in an initial stage of assembly.

(23) FIG. **23** is an enlarged front elevational view of the retainer, receiver and insert with portions broken away, similar to what is shown in FIG. **22**, showing the insert in a subsequent stage of assembly, resilient arms of the insert pressing against inner surfaces of the receiver.

(24) FIG. **24** is a reduced front elevational view with portions broken away, similar to FIG. **23**, and further showing an enlarged and partial shank of FIG. **1** in a first stage of assembly with the retainer, a hemisphere of the shank head and a vertebra portion both being shown in phantom.

(25) FIG. **25** is a partial front elevational view with portions broken away, similar to FIG. **24**, showing the retainer in an expanded state about a mid-portion of the shank head, the head hemisphere shown in phantom.

(26) FIG. **26** is a partial front elevational view with portions broken away, similar to FIG. **25**, the shank upper portion or head fully captured by the retainer.

(27) FIG. **27** is another partial front elevational view with portions broken away, similar to FIG. **26**, the shank upper portion or head being in frictional engagement with lower panels of the insert.

(28) FIG. **28** is an enlarged and partial cross-sectional view taken along the line **28-28** of FIG. **27**.

(29) FIG. **29** is a partial front elevational view with portions broken away, similar to FIG. **27**, the shank upper portion and retainer being shown pulled down into a seated position within the lower receiver cavity, the insert arms in a substantially neutral state, capturing the insert below a ledge of the receiver.
(30) FIG. **30** is a partial front elevational view with portions broken away of all of the components shown in FIG. **1**, the assembly as in FIG. **29** being shown in a stage of assembly with the rod and closure top.
(31) FIG. **31** is an enlarged and partial front elevational view with portions broken away, similar to FIG. **30**, shown in a final locking position.
(32) FIG. **32** is another enlarged and partial front elevational view with portions broken away, similar to FIG. **31**.
(33) FIG. **33** is an enlarged cross-sectional view taken along the line **33-33** of FIG. **31**.
(34) FIG. **34** is an enlarged perspective view of an alternative locking insert for use with the assembly of FIG. **1** in lieu of the insert shown in FIG. **1**.
(35) FIG. **35** is a top plan view of the insert of FIG. **34**.
(36) FIG. **36** is a front elevational view of the insert of FIG. **34**.
(37) FIG. **37** is an enlarged front elevational view with portions broken away of the receiver and retainer of FIG. **1** and the insert of FIG. **34** in reduced front elevation, the insert shown captured within the receiver and in an un-locked shipping position.
(38) FIG. **38** is a partial front elevational view of the shank, receiver, retainer, rod and closure of FIG. **1**, with portions broken away and assembled with the locking insert as shown in FIG. **37**, but in a locked stage of assembly.
(39) FIG. **39** is an enlarged and partial front elevational view, similar to FIG. **38**, illustrating the interference locking of the insert against the receiver.
(40) FIG. **40** is an enlarged and partial cross-sectional view taken along the line **40-40** of FIG. **38**.
(41) FIG. **41** is an enlarged and partial front elevational view with portions broken away, similar to FIG. **39**, showing the shank, retainer, insert and receiver remaining in a locked position after removal of the rod and closure top of FIG. **1** and further showing, in exploded view, an alternative deformable rod and cooperating alternative closure top.
(42) FIG. **42** is a partial front elevational view with portions broken away, similar to FIG. **41**, showing the alternative rod and closure top fixed to the remainder of the assembly.
(43) FIG. **43** is a reduced and partial front elevational view with portions broken away of the assembly of FIG. **42** without the alternative rod and closure top, and further showing unlocking of the insert from the receiver with a two-piece tool having an inner insert engaging portion and an outer tubular holding portion.
(44) FIG. **44** is a reduced and partial front elevational view of the two-piece tool of FIG. **43**, holding prongs of the inner insert engaging portion being shown in phantom.
(45) FIG. **45** is an enlarged and partial perspective view of the inner insert engaging portion of the tool shown in FIG. **44** with portions broken away to show the detail thereof.
(46) FIG. **46** is an enlarged and partial perspective view of the assembly of FIG. **30**, but shown with the shank being at an angle with respect to the receiver and further showing an alternative locking tool for independently locking the shank with respect to the receiver when the closure top and rod are in a loose, unlocked relationship with the receiver as shown.
(47) FIG. **47** is a partial perspective view of a portion of the locking tool of FIG. **46**.
(48) FIG. **48** is an enlarged and partial front elevational view of the assembly and locking tool of FIG. **46** with portions broken away to show the detail thereof.
(49) FIG. **49** is an exploded perspective view of another embodiment of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, a retainer in the form of an open ring and a friction fit crown compression insert, further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

(50) FIG. **50** is an enlarged top plan view of the shank of FIG. **49**.
(51) FIG. **51** is reduced cross-sectional view taken along the line **51-51** of FIG. **50**.
(52) FIG. **52** is an enlarged perspective view of the receiver of FIG. **49**.
(53) FIG. **53** is a side elevational view of the receiver of FIG. **52**.
(54) FIG. **54** is a top plan view of the receiver of FIG. **52**.
(55) FIG. **55** is a bottom plan view of the receiver of FIG. **52**.
(56) FIG. **56** is an enlarged cross-sectional view taken along the line **56-56** of FIG. **54**.
(57) FIG. **57** is an enlarged cross-sectional view taken along the line **57-57** of FIG. **54**.
(58) FIG. **58** is an enlarged perspective view of the retainer of FIG. **49**.
(59) FIG. **59** is a front elevational view of the retainer of FIG. **58**.
(60) FIG. **60** is a top plan view of the retainer of FIG. **58**.
(61) FIG. **61** is a bottom plan view of the retainer of FIG. **58**.
(62) FIG. **62** is an enlarged cross-sectional view taken along the line **62-62** of FIG. **60**.
(63) FIG. **63** is an enlarged perspective view of the insert of FIG. **49**.
(64) FIG. **64** is a front elevational view of the insert of FIG. **63**.
(65) FIG. **65** is another perspective view of the insert of FIG. **63**.
(66) FIG. **66** is a side elevational view of the insert of FIG. **63**.
(67) FIG. **67** is a top plan view of the insert of FIG. **63**.
(68) FIG. **68** is a bottom plan view of the insert of FIG. **63**.
(69) FIG. **69** is an enlarged cross-sectional view taken along the line **69-69** of FIG. **67**.
(70) FIG. **70** is an enlarged cross-sectional view taken along the line **70-70** of FIG. **67**.
(71) FIG. **71** is an enlarged front elevational view of the retainer and receiver of FIG. **49** with portions of the receiver broken away to show the detail thereof and intermediate positions of the retainer while being downloaded into the receiver being shown in phantom.
(72) FIG. **72** is a front elevational view with portions broken away, similar to FIG. **71**, further showing the insert of FIG. **49** in enlarged side elevation, with an early stage of assembly of the insert being shown in phantom.
(73) FIG. **73** is a front elevational view with portions broken away, similar to FIG. **72**, showing the insert rotated within the receiver during an assembly stage subsequent to that shown in FIG. **72**.
(74) FIG. **74** is an enlarged perspective view with portions broken away of the assembly shown in FIG. **73** and further showing a subsequent step of crimping a portion of the receiver against the insert.
(75) FIG. **75** is an enlarged side elevational view of the assembly shown in FIG. **74**.
(76) FIG. **76** is a reduced front elevational view with portions broken away, similar to FIG. **73** and with the crimping of FIGS. **74** and **75** and further showing an alternative assembly stage with the shank of FIG. **49** shown in partial front elevation in which the shank is first implanted in a vertebra, shown in phantom, followed by assembly with the receiver, retainer and insert.
(77) FIG. **77** is an enlarged and partial front elevational view with portions broken away, similar to FIG. **76** showing the shank (not implanted in a vertebra) in a stage of assembly with the retainer, the retainer being pushed up into engagement with the insert.
(78) FIG. **78** is an enlarged partial front elevational view with portions broken away, similar to FIG. **77**, showing the retainer in an expanded state about an upper portion of the shank, the shank upper portion in a stage of assembly with the insert.
(79) FIG. **79** is a reduced partial front elevational view of the assembly as shown in FIG. **78**, with further portions broken away to show the stage of assembly between the shank upper portion, the retainer and the insert.
(80) FIG. **80** is a reduced partial front elevational view with portions broken away, similar to FIG. **78**, the shank upper portion in frictional engagement with the insert and the retainer in a substantially neutral state.
(81) FIG. **81** is an enlarged partial front elevational view of the assembly as shown in FIG. **80**, with

further portions broken away to show the engagement between the shank upper portion and the insert.
(82) FIG. **82** is a reduced partial front elevational view with portions broken away, similar to FIGS. **80** and **81**, showing the assembly in a locked position with the rod and closure top of FIG. **49**, also shown in front elevation with portions broken away.
(83) FIG. **83** is a reduced and partial side elevational view of the assembly of FIG. **82**.
(84) FIG. **84** is an enlarged perspective view of an alternative locking insert according to the invention for use in the assembly of FIG. **49** in lieu of the insert shown in FIG. **49**.
(85) FIG. **85** is an enlarged side elevational view of the insert of FIG. **84** with portions broken away to show the detail thereof.
(86) FIG. **86** is an enlarged front elevational view of the insert of FIG. **84** with portions broken away to show the detail thereof.
(87) FIG. **87** is an enlarged front elevational view of the receiver and retainer of FIG. **49** shown assembled with the insert of FIG. **84**, also in front elevation, with portions broken away to show the detail thereof.
(88) FIG. **88** is a reduced side elevational view of the assembly of FIG. **87**.
(89) FIG. **89** is a reduced front elevational view of the receiver (with portions broken away), retainer and insert of FIG. **87** further shown assembled with a shank of FIG. **49**, shown in partial front elevation, and in a stage of assembly with the rod and closure top of FIG. **49**, also shown in front elevation.
(90) FIG. **90** is an enlarged and partial front elevational view with portions broken away, similar to FIG. **89** but showing the insert in locked assembly with the receiver, retainer, rod and closure top.
(91) FIG. **91** is a reduced partial front elevational view with portions broken away, similar to FIG. **90**, but shown with the rod and closure top removed, the locking insert remaining in locked relation with respect to the receiver, and further being shown with an alternative deformable rod and cooperating closure top, shown in exploded view.
(92) FIG. **92** is a partial front elevational view with portions broken away, similar to FIG. **91** showing the alternative deformable rod and closure top in locked relationship with the rest of the assembly.
(93) FIG. **93** is a reduced and partial front elevational view with portions broken away of the assembly of FIG. **91** without the alternative rod and closure top, and further showing unlocking of the insert from the receiver with a two-piece tool having an inner insert engaging portion and an outer tubular holding portion.
(94) FIG. **94** is a reduced and partial front elevational view of the two-piece tool of FIG. **93**, holding prongs of the inner insert engaging portion being shown in phantom.
(95) FIG. **95** is an enlarged and partial perspective view of the inner insert engaging portion of the tool shown in FIG. **94** with portions broken away to show the detail thereof.
(96) FIG. **96** is an enlarged and partial perspective view of an assembly identical to that shown in FIG. **90** with the exception of the shank being at an angle with respect to the receiver and further showing an alternative locking tool for independently locking the shank with respect to the receiver when the closure top and rod are in a loose, unlocked relationship with the receiver.
(97) FIG. **97** is a partial perspective view of a portion of the locking tool of FIG. **96**.
(98) FIG. **98** is an enlarged and partial front elevational view of the assembly and locking tool of FIG. **96** with portions broken away to show the detail thereof.
DETAILED DESCRIPTION OF THE INVENTION
(99) As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present invention in

virtually any appropriately detailed structure. It is also noted that any reference to the words top, bottom, up and down, and the like, in this application refers to the alignment shown in the various drawings, as well as the normal connotations applied to such devices, and is not intended to restrict positioning of the bone attachment structures in actual use.

(100) With reference to FIGS. **1-33** the reference number **1** generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **1** includes a shank **4**, that further includes a body **6** integral with an upwardly extending upper portion or head structure **8**; a receiver **10**; an open ring retainer **12**, and a compression or pressure insert **14** having structure for friction fit non-locking engagement with the shank head **8**. The receiver **10**, retainer **12** and compression insert **14** are initially assembled and may be further assembled with the shank **4** either prior or subsequent to implantation of the shank body **6** into a vertebra **17** (see FIG. **24**), as will be described in greater detail below. FIGS. **1** and **31-33** further show a closure structure **18** for capturing a longitudinal connecting member, for example, a rod **21** which in turn engages the compression insert **14** that presses against the shank upper portion **8** into fixed frictional contact with the retainer **12**, so as to capture, and fix the longitudinal connecting member **21** within the receiver **10** and thus fix the member **21** relative to the vertebra **17**. The illustrated rod **21** is hard, stiff, non-elastic and cylindrical, having an outer cylindrical surface **22**. It is foreseen that in other embodiments, the rod **21** may be elastic, deformable and/or of different materials and cross-sectional geometries. The receiver **10** and the shank **4** cooperate in such a manner that the receiver **10** and the shank **4** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **10** with the shank **4** until both are locked or fixed relative to each other near the end of an implantation procedure.

(101) The shank **4**, best illustrated in FIGS. **1-3**, is elongate, with the shank body **6** having a helically wound bone implantable thread **24** (single or dual lead thread form and different thread types) extending from near a neck **26** located adjacent to the upper portion or head **8**, to a tip **28** of the body **6** and extending radially outwardly therefrom. During use, the body **6** utilizing the thread **24** for gripping and advancement is implanted into the vertebra **17** leading with the tip **28** and driven down into the vertebra with an installation or driving tool (not shown), so as to be implanted in the vertebra to a location at or near the neck **26**, as more fully described in the paragraphs below. The shank **4** has an elongate axis of rotation generally identified by the reference letter A.

(102) The neck **26** extends axially upward from the shank body **6**. The neck **26** may be of the same or is typically of a slightly reduced radius as compared to an adjacent upper end or top **32** of the body **6** where the thread **24** terminates. Further extending axially and outwardly from the neck **26** is the shank upper portion or head **8** that provides a connective or capture apparatus disposed at a distance from the upper end **32** and thus at a distance from the vertebra **17** when the body **6** is implanted in such vertebra.

(103) The shank upper portion **8** is configured for a pivotable connection between the shank **4** and the retainer **12** and receiver **10** prior to fixing of the shank **4** in a desired position with respect to the receiver **10**. The shank upper portion **8** has an outer, convex and substantially spherical surface **34** that extends outwardly and upwardly from the neck **26** that in some embodiments terminates at a substantially planar top or rim surface **38**. In the illustrated embodiment, a frusto-conical surface **39** extends from the spherical surface **34** to the top surface **38**, providing additional clearance during pivoting of the shank with respect to the receiver **10** and the insert **14**. The spherical surface **34** has an outer radius configured for temporary frictional, non-floppy, sliding cooperation with panels of the insert **14** having concave or flat surfaces, as well as ultimate frictional engagement with the insert **14** at upper inner stepped surface thereof, as will be discussed more fully in the paragraphs below. The shank top surface **38** is substantially perpendicular to the axis A. The spherical surface **34** shown in the present embodiment is substantially smooth, but in some embodiments may include a roughening or other surface treatment and is sized and shaped for cooperation and

ultimate frictional engagement with the compression insert **14** as well as ultimate frictional engagement with the retainer **12**. The shank spherical surface **34** is locked into place exclusively by the insert **14** and the retainer **12** and not by inner surfaces defining the receiver cavity.

(104) A counter sunk substantially planar base or stepped seating surface **45** partially defines an internal drive feature or imprint **46**. The illustrated internal drive feature **46** is an aperture formed in the top surface **38** and has a star shape designed to receive a tool (not shown) of an Allen wrench type, into the aperture for rotating and driving the bone screw shank **4**. It is foreseen that such an internal tool engagement structure may take a variety of tool-engaging forms and may include one or more apertures of various shapes, such as a pair of spaced apart apertures or a multi-lobular or hex-shaped aperture. The seat or base surfaces **45** of the drive feature **46** are disposed substantially perpendicular to the axis A with the drive feature **46** otherwise being coaxial with the axis A. As illustrated in FIGS. **2** and **3**, the drive seat **45** may include beveled or stepped surfaces that may further enhance gripping with the driving tool. In operation, a driving tool (not shown) is received in the internal drive feature **46**, being seated at the base **45** and engaging the faces of the drive feature **46** for both driving and rotating the shank body **6** into the vertebra **17**, either before the shank **4** is attached to the receiver **10** as shown, for example, in FIG. **24**, or after the shank **4** is attached to the receiver **10**, with the shank body **6** being driven into the vertebra **17** with the driving tool extending into the receiver **10**.

(105) The shank **4** shown in the drawings is cannulated, having a small central bore **50** extending an entire length of the shank **4** along the axis A. The bore **50** is defined by an inner cylindrical wall of the shank **4** and has a circular opening at the shank tip **28** and an upper opening communicating with the external drive **46** at the driving seat **45**. The bore **50** is coaxial with the threaded body **6** and the upper portion **8**. The bore **50** provides a passage through the shank **4** interior for a length of wire (not shown) inserted into the vertebra **17** prior to the insertion of the shank body **6**, the wire providing a guide for insertion of the shank body **6** into the vertebra **17**. It is foreseen that the shank could be solid and made of different materials, including metal and non-metals.

(106) To provide a biologically active interface with the bone, the threaded shank body **6** may be coated, perforated, made porous or otherwise treated. The treatment may include, but is not limited to a plasma spray coating or other type of coating of a metal or, for example, a calcium phosphate; or a roughening, perforation or indentation in the shank surface, such as by sputtering, sand blasting or acid etching, that allows for bony ingrowth or ongrowth. Certain metal coatings act as a scaffold for bone ingrowth. Bio-ceramic calcium phosphate coatings include, but are not limited to: alpha-tri-calcium phosphate and beta-tri-calcium phosphate (Ca.sub.3(PO.sub.4).sub.2, tetra-calcium phosphate (Ca.sub.4P.sub.2O.sub.9), amorphous calcium phosphate and hydroxyapatite (Ca.sub.10(PO.sub.4).sub.6(OH).sub.2). Coating with hydroxyapatite, for example, is desirable as hydroxyapatite is chemically similar to bone with respect to mineral content and has been identified as being bioactive and thus not only supportive of bone ingrowth, but actively taking part in bone bonding.

(107) With particular reference to FIGS. **1** and **4-9**, the receiver **10** has a generally U-shaped appearance with a partially discontinuous, partially faceted and partially curved outer profile and partially cylindrical inner and outer profiles. The receiver **10** has an axis of rotation B that is shown in FIG. **1** as being aligned with and the same as the axis of rotation A of the shank **4**, such orientation being desirable, but not required during assembly of the receiver **10** with the shank **4**. After the receiver **10** is pivotally attached to the shank **4**, either before or after the shank **4** is implanted in a vertebra **17**, the axis B is typically disposed at an angle with respect to the axis A, as shown, for example, in FIG. **46** that illustrates the assembly **1** with a manipulation and locking tool.

(108) The receiver **10** includes a curvate lower base portion **60** defining a bore or inner cavity, generally **61**, the base **60** being integral with a pair of opposed upstanding arms **62** forming a cradle and defining a channel **64** between the arms **62** with an upper opening, generally **66**, and a substantially planar lower channel portion or seat **68**, the channel **64** having a width for operably

receiving the rod **21** or portion of another longitudinal connector between the arms **62**, as well as closely receiving laterally extending portions of the insert **14**, the channel **64** communicating with the base cavity **61**. Inner opposed substantially planar perimeter arm surfaces **69** partially define the channel **64** and are located on either side of each arm interior substantially cylindrical surfaces generally **70**. Lower opposed surface portions **71** of the arm surfaces **69** terminate at the lower seat **68**. The arm interior surfaces **70**, each include various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **72** located adjacent top surfaces **73** of each of the arms **62**. In the illustrated embodiment, the guide and advancement structure **72** is a partial helically wound interlocking flange form configured to mate under rotation with a similar structure on the closure structure **18**, as described more fully below. However, it is foreseen that for certain embodiments of the invention, the guide and advancement structure **72** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-thread-like helically wound discontinuous advancement structures, for operably guiding under rotation and advancing the closure structure **18** downward between the arms **62**, as well as eventual torquing when the closure structure **18** abuts against the rod **21** or other longitudinal connecting member. It is foreseen that the arms **62** could have break-off extensions.

(109) An opposed pair of upper rounded off triangular or delta-shaped tool receiving and engaging apertures **74**, each having a through bore formed by an upper arched surface **75** and a substantially planar bottom surface **75′**, are formed on outer surfaces **76** of the arms **62**. Each through bore surface **75** and **75′** extends through the arm inner surface **70**. The apertures **74** with through bore portions **75** and **75′** are sized and shaped for receiving locking, unlocking and other manipulation tools and in the illustrated embodiment, receives the retainer ring **12** (as shown in phantom in FIG. **21**) during top loading of the retainer **12** into the receiver **10** will be described in greater detail below. Each aperture **74** further includes a sloping tool alignment surface **77** that surrounds the arched bore portion **75** and does not extend completely through the respective arm **62**. It is noted that the receiver **10** is an integral structure and devoid of any spring tabs or collet-like structures. As will be discussed in greater detail below, the geometry of the insert **14** that extends outwardly into the receiver channel **64** at the perimeter arms surfaces **69** prohibits the insert **14** from rotating during assembly and thus prohibits any misalignments with the receiver **10** and the rod **21** or other longitudinal connecting member that sometimes occur with other types of compression inserts. The apertures **74** and additional tool receiving apertures or grooves (not shown) may be used for holding the receiver **10** during assembly with the insert **14**, the retainer **12** and the shank **4**; during the implantation of the shank body **6** into a vertebra when the shank is pre-assembled with the receiver **10**; during assembly of the bone anchor assembly **1** with the rod **21** and the closure structure **18**; and during lock and release adjustment of some inserts according to the invention with respect to the receiver **10**, either into or out of frictional engagement with the inner surfaces of the receiver **10** as will be described in greater detail below. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **62**.

(110) Returning to the interior surface **70** of the receiver arms **62**, located below the guide and advancement structure **72** is a discontinuous cylindrical surface **80** partially defining a run-out feature for the guide and advancement structure **72**. The cylindrical surface **80** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure **72**. Moving downwardly, in a direction toward the base **60**, following the cylindrical surface **80** of each arm is a cylindrical surface **82** located below an annular run-out seat or surface **83** that extends inwardly toward the axis B and runs perpendicular or somewhat obliquely towards the axis B. The surface **82** has a diameter smaller than the diameter of the surface **80**. The surface **82** is sized and shaped to initially closely receive and frictionally hold a portion of the insert **14** as will be described in greater detail below. Located below the surface **82** is a discontinuous annular surface or narrow ledge **84** that is disposed substantially perpendicular to the axis B. A partially discontinuous

cylindrical surface **86** is located on each arm below and adjacent to the ledge surface **84**. The surface **86** has a diameter larger than the diameter of the surface **82**. A portion of the aperture surface **75** is adjacent to the cylindrical surface **86** at each arm, the surface **86** otherwise terminating at a lower ledge **87**. A partially discontinuous cylindrical surface **88** is located on each arm below and adjacent to the lower ledge **87**. A diameter of the surface **88** is larger than the diameter of the surface **86**. The surface **88** terminates at a lip **89** that extends inwardly toward the receiver axis B. Located below the lip **89** is a partially discontinuous cylindrical surface **90**. The surface **90** also partially defines the inner cavity **61** generally below the apertures **74** at the surface **75′** and below the channel seat **68**. The cylindrical surface **90** has a diameter slightly larger than the diameter of the discontinuous surface **88**. The surface **90** terminates at a cavity lower ledge surface **91** that extends outwardly away from the axis B and that may be perpendicular to the axis B, but is illustrated as a frusto-conical surface that extends both downwardly and outwardly away from the axis B. The through bores **75** of the apertures **74** each extend through the arms at the surfaces **86**, **88** and **90** with the sloping tool engagement walls **77** extending substantially on either side of each bore surface **75** and formed in the arm outer surfaces **76** at a location primarily opposite the inner surfaces **86** and **88**.

(111) With particular reference to FIGS. **1**,**5**, **6** and **8**, returning to the substantially planar peripheral surfaces **69**, each arm **62** includes a pair of projecting ridges or stops **92**, located on each surface **69**, for a total of four stops **92** that are located near the annular surface **87** and extend from front and back surfaces or arm faces **94** to the cylindrical surface **88**. The stops **92** of one arm **62** directly face the opposing pair of stops **92** on the other arm **62**, each stop **92** projecting outwardly away from the respective planar surface **69**. The illustrated stops **92** are elongate and run in a direction perpendicular to the axis B. As will be described in greater detail below, the stops **92** cooperate with surfaces of the insert **14** to retain the insert **14** within the channel **64** of the receiver **10**. In the illustrated embodiment, each stop **92** includes a bottom surface or ledge **95** adjacent to a partially planar and partially curved surface **96**. A planar portion of the surface **96** located directly beneath the stop **92** is in line with or may be slightly inset from the surface **69**. Each set of opposed surfaces **96** curve toward one another and terminate at the respective adjacent lower surface portions **71**. An edge **97** defines a juncture of each curved surface **96** and the respective adjacent lower surface portion **71**. A first width measured between opposing surface portions **71** is smaller than a second width measured between opposed surfaces **69** located between the stops **92** and arm top surfaces **73**, providing opposed planar locking interference fit surfaces for the insert **14′** as will be described in greater detail below. The insert **14** is sized and shaped to be closely received but slidable between the surfaces **71**.

(112) Returning to FIGS. **8** and **9**, the annular or frusto-conical surface **91** partially defines the base cavity **61** and is located below and adjacent to the cylindrical surface **90**. Another cylindrical surface **99** is located below and adjacent to the surface **91**. The surface **99** also defines a portion of the base cavity **61**. The cylindrical surface **99** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded retainer ring **12**. The surfaces **91** and **99** define a circumferential recess that is sized and shaped to receive the retainer **12** as it expands around the shank upper portion **8** as the shank **8** moves upwardly through the receiver base and toward the channel **64** during assembly. It is foreseen that the recess could be tapered or conical in configuration. A cylindrical surface **101** located below the cylindrical surface **99** is sized and shaped to closely receive and surround a lower portion of the retainer **12** when the retainer is in a substantially neutral position as shown in FIG. **37**, for example. Thus, the cylindrical surface **101** has a diameter smaller than the diameter of the cylindrical surface **99** that defines the expansion area or expansion chamber for the retainer **12**. The surface **101** is joined or connected to the surface **99** by one or more beveled, curved or conical surfaces **102**. The surfaces **102** allow for sliding and neutral or nominal positioning of the retainer **12** into the space defined by the surface **101** and ultimate seating of the retainer **12** on a lower substantially horizontal annular surface **104** located

below and adjacent to the cylindrical surface **101**. The annular surface **104** is disposed substantially perpendicular to the axis B.

(113) Located below and adjacent to the annular seating surface **104** is another substantially cylindrical surface **106** that communicates with a beveled or flared bottom opening surface **107**, the surface **107** communicating with an exterior base surface **108** of the base **60**, defining a lower opening, generally **110**, into the base cavity **61** of the receiver **10**.

(114) With particular reference to FIGS. **1** and **10-13**, the lower open or split retainer **12**, that operates to capture the shank upper portion **8** within the receiver **10**, has a central axis that is operationally the same as the axis B associated with the receiver **10** when the shank upper portion **8** and the retainer **12** are installed within the receiver **10**. The retainer ring **12** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **12** may be expanded during various steps of assembly as will be described in greater detail below. The retainer **12** has a central channel or hollow through bore, generally **121**, that passes entirely through the ring **12** from a top surface **122** to a bottom surface **124** thereof. Surfaces that define the channel or bore **121** include a discontinuous inner cylindrical surface **125** adjacent the top surface **122** and a discontinuous frusto-conical surface **127** adjacent the surface **125**, both surfaces coaxial when the retainer **12** is in a neutral non-compressed, non-expanded orientation. An edge **128** is defined by the juncture of the top surface **122** and the cylindrical surface **125**. As shown, for example, in FIG. **31**, the shank upper portion **8** ultimately frictionally engages the retainer **12** at the edge **128** when the assembly **1** is locked into a final position. The retainer **12** further includes an outer cylindrical surface **130** located adjacent the top surface **122** and an outer beveled or frusto-conical surface **132** adjacent the bottom surface **124**. The surface **130** is oriented parallel to the central axis of the retainer **12**. In some embodiments of the invention, spaced notches (not shown) may be formed in the cylindrical surface **130** to receive a holding and manipulation tool (not shown). In some embodiments further notches on inner or outer surfaces of the retainer may be made to evenly distribute stress across the entire retainer **12** during expansion thereof.

(115) The resilient retainer **12** further includes first and second end surfaces, **134** and **135** disposed in spaced relation to one another when the retainer is in a neutral non-compressed state. The surface **134** and **135** may also be touching when the retainer is in a neutral state. Both end surfaces **134** and **135** are disposed substantially perpendicular to the top surface **122** and the bottom surface **124**. A width X between the surfaces **134** and **135** is very narrow (slit may be made by EDM process) to provide stability to the retainer **12** during operation. Because the retainer **12** is top loadable in a neutral state and the retainer **12** does not need to be compressed to fit within the receiver cavity **61**, the width X may be much smaller than might be required for a bottom loaded compressible retainer ring. The gap X functions only in expansion to allow the retainer **12** to expand about the shank upper portion **8**. This results in a stronger retainer that provides more surface contact with the shank upper portion **8** upon locking, resulting in a sturdier connection with less likelihood of failure than a retainer ring having a greater gap. Furthermore, because the retainer **12** is only expanded and never compressed inwardly, the retainer **12** does not undergo the mechanical stress that typically is placed on spring ring type retainers known in the prior art that are both compressed inwardly and expanded outwardly during assembly.

(116) It is foreseen that in some embodiments of the invention, the retainer **12** inner surfaces may include a roughening or additional material to increase the friction fit against the shank upper portion **8** prior to lock down by the rod **21** or other longitudinal connecting member. Also, the embodiment shown in FIGS. **10-13** illustrates the surfaces **134** and **135** as substantially parallel to the central axis of the retainer, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle.

(117) With particular reference to FIGS. **1** and **14-20**, the friction fit compression insert **14** is illustrated that is sized and shaped to be received by and down-loaded into the receiver **10** at the upper opening **66**. The compression insert **14** has an operational central axis that is the same as the

central axis B of the receiver **10**. In operation, the insert advantageously frictionally engages the bone screw shank upper portion **8**, allowing for un-locked but non-floppy placement of the angle of the shank **4** with respect to the receiver **10** during surgery prior to locking of the shank with respect to the receiver near the end of the procedure. As will be described in greater detail below with respect to the insert **14′** illustrated in FIGS. **34-40**, in some embodiments of the invention, the insert that has locked the shank **4** in a desired angular position with respect to the receiver **10**, by, for example, compression from the rod **21** and closure top **18**, is also forced into an interference fit engagement with the receiver **10** at the pair of opposed receiver planar arm surfaces **71** thereof and thus is capable of retaining the shank **6** in a locked position even if the rod **21** and closure top **18** are removed. Such locked position may also be released by the surgeon if desired. The non-locking insert **14** as well as the locking insert **14′** are preferably made from a solid resilient material, such as a stainless steel or titanium alloy, to provide for friction fit panels and also so that arm portions of the insert may be pinched or pressed toward one another in some embodiments, such portions pressing outwardly against the receiver **10** during shipping and early stages of assembly.

(118) The non-locking compression insert **14** includes a body **150** having a partially outer cylindrical surface and an inner U-shaped surface, the insert having opposed ends, generally **151**, the insert **14** being sized and shaped to extend completely through the U-shaped channel **64** between the opposed front and back surfaces or faces **94** of the arms **62** so as to cooperate with the receiver arm side surfaces **69**, the stops **92**, the surfaces **96** and **71** below the stops **92** and the channel seat **68**. A U-shaped channel surface or saddle **153** formed in the body **150** also extends between the insert ends **151** and when the insert **14** is assembled with the receiver **10**, the saddle **153** substantially aligns with the receiver channel **64**. The saddle **153** is formed by the insert body **150** and by two upstanding arms **156** and is sized and shaped to closely receive the rod **21** or other longitudinal connecting member. It is foreseen that an alternative insert embodiment may be configured to include planar holding surfaces that closely hold a square or rectangular bar as well as hold a cylindrical rod-shaped, cord, or sleeved cord longitudinal connecting member.

(119) The insert **14** body **150** is thus integral with the pair of upstanding arms **156** at an upper end thereof and is also integral with a downwardly extending super structure illustrated as an opposed pair of crown collet extensions **158** at a lower end thereof, each super structure extension **158** terminating at a slotted bottom surface **159**. A bore, generally **160**, is disposed primarily within and through the insert body **150** that runs along the axis B and communicates with the U-shaped channel formed by the saddle **153** and upstanding arms **156** and also runs between the collet extensions **158**. The bore **160** is sized and shaped to provide space and clearance for shank driving and other manipulation tools.

(120) The arms **156** that are disposed on either side of the saddle **153** extend upwardly therefrom and are sized and configured for ultimate placement beneath and spaced from the closure top **118** within the receiver cylindrical surfaces **86** and **88**. Inner upper arm surfaces **157** extend upwardly and slightly outwardly from the remainder of the U-shaped saddle **153**. The arms **156** are also sized and shaped for resilient temporary placement at the receiver cylindrical surface **82**. The arms **156** include outer lower convex surfaces **162** that are illustrated as partially cylindrical, outer upper curved surfaces **163** adjacent the surfaces **162**, the surfaces **163** being curved and convex and also flaring outwardly from either side of the body **150**. The surfaces **163** are adjacent planar top surfaces **164** that are ultimately positioned in spaced relation with the closure top **18**, so that the closure top **18** frictionally engages the rod **21** only, pressing the rod **21** downwardly against the insert saddle **153**, the shank **4** upper portion **8** then pressing against the retainer **12** to lock the polyaxial mechanism of the bone screw assembly **1** at a desired angle. Each of the top surfaces **164** slopes upwardly and away from the adjacent saddle surface **157**.

(121) Each partially cylindrical surface **162** located below the respective flared surface **163** extends from the surface **163** to a partially annular lower or bottom surface **165** of the insert body **150**. Each surface **162** is sized and shaped to generally fit within the receiver arms inner arm surfaces,

generally **70**. It is foreseen that in some embodiments of the invention, the arms **156** may be extended and the top surfaces not sloped and the closure top configured such that the arms and, more specifically, the arm top surfaces ultimately directly engage the closure top **18** for locking of the polyaxial mechanism, for example, when the rod **21** is made from a deformable material. The arm outer surfaces **162** further include notches or grooves formed thereon for receiving and engaging locking tools. Specifically, in the illustrated embodiment, each surface **162** has a v-notch or recess for receiving tooling, the notch defined by an upper sloping surface **167** and intersecting a lower planar surface **168** disposed substantially perpendicular to the central operational axis of the insert **14**. The surfaces **167** and **168** cooperate and align with the respective receiver aperture through bore **75**, surface, and surface **75′** when the insert **14** is captured and operationally positioned within the receiver **10** as will be described in greater detail below. It is also foreseen that the arms **163** can extend upwardly and not be flared or tapered outwardly in some embodiments.

(122) The insert **14** extends from the substantially cylindrical outer arms surfaces **162** equally outwardly to each end **151**. Substantially planar outer side surfaces **170** extend from each arm surface **162** to a substantially planar surface **171** disposed perpendicular thereto, the surfaces **171** substantially defining each of the ends **151**. Each end surface **171** is adjacent to a lower extension surface **172** that runs substantially parallel to the surface **165** and extends inwardly toward the insert body **150**. Adjacent to each side surface **170** is a substantially planar upper or top surface **173** running from one of the arms **156** to each of the end surfaces **171**. Each of the surfaces **170** form a narrow outer strip and are adjacent and perpendicular to a lower narrow ledge **174**. The ledges **174** run parallel to the upper surfaces **173**. An inset planar surface **175** is adjacent to each lower ledge surface **174** and runs parallel to the respective outer planar side surface **170**. A width between opposing surfaces **175** is sized such that the surfaces **175** are slidingly received between the opposed receiver lower arm surfaces **71**. In other embodiments of the invention, a width between the surfaces **175** may be enlarged such that the surfaces **175** must be forced downwardly between the planar surfaces **71** to provide a locking non-contractile type of interference fit of the insert against the receiver and thus lock the polyaxial mechanism of the bone screw assembly as will be described below with respect to the insert **14′**. The surfaces **175** terminate at planar end surfaces **177** and lower extension surfaces **178**. A portion of each surface **175** extends all the way to the planar end surface **171**.

(123) The insert bore, generally **160**, is substantially defined at the body **150** by an inner substantially cylindrical surface **180** that communicates with the saddle **153** and also communicates with a lower concave substantially spherical surface **181** having a radius the same or substantially similar to a radius of the surface **34** of the shank upper portion or head **8**. A portion of the surface **181** terminates at the body lower surface **165**. The surface **181** also defines inner surfaces of the crown collet extensions **158**. Located along the spherical surface **181** between the cylindrical surface **180** and the partially annular lower body surface **165** is a shank gripping surface portion **182**. The gripping surface portion **182** includes one or more stepped surfaces or ridges sized and shaped to grip and penetrate into the shank head **8** when the insert **14** is locked against the head surface **34**. It is foreseen that the stepped surface portion **182** may include greater or fewer number of stepped surfaces and cover greater or less surface area of the spherical surface **181**. It is foreseen that the shank gripping surface portion **182** and also the spherical surface **181** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the shank upper portion **8**.

(124) The two collet extensions **158** that generally extend in a direction opposite to the two arms **156** and have the discontinuous inner spherical surface **181**, also include through slits or slots **183** running substantially vertically from adjacent the shank gripping surface portion **182** through the bottom surfaces **159**. The illustrated embodiment includes one slot **183** centrally located in each extension **158**. It is foreseen that other embodiments of the invention may include more or fewer slots **183** or no slots. The slots **183** substantially equally partition each of the extensions **158**,

forming four distinct resilient, partially spherical fingers, tab or panels **196** that extend from the shank gripping portion **182** to the bottom surface **159**. In other words, the discontinuous inner spherical surface **181** is further separated into four sections or panels **184**, each having the discontinuous partially inner spherical surface **181** and having an outer surface **185** that is substantially cylindrical in form. The panels **184** are sized and shaped to resiliently expand about the spherical surface **34** of the shank upper portion **8** and then snap on and frictionally grip the surface **34**. In the illustrated embodiment, the spherical surface **181** is designed such that the gripping tabs or panels **184** have a neutral or non-expanded radius that is the same or slightly smaller than a radius of the shank surface **34** so that when the tabs or panels **184** are gripping the surface **34**, the insert **14** collet extension portion **138** is in a neutral or slightly expanded state. In other embodiments, the non-expanded radius is the same or larger than a radius of the shank surface. The contacting surface area between the shank and the insert is sufficient to provide a non-floppy frictional fit in such instances. Furthermore, the shank surface **34** and/or the spherical surface **181** may include a roughened or grooved surface feature to provide for a frictional fit between the shank and the insert. In other embodiments, the resilient panels **184** having a slightly larger pre-assembly radius than the shank surface **34** and may be bent inwardly to result in a tighter frictional fit with the shank surface. When the shank **4** is locked into position by a rod **21** or other connecting member being pressed downwardly on the insert seat **153** by the closure top **18**, the insert **14** shank gripping portion **182** that is initially slidable along the shank surface **34** then digs or penetrates into the surface **34** and thus securely fixes the shank upper portion **8** to the insert at the portion **182**. It is foreseen that the spherical surfaces **181** could be flat or planar in some embodiments.

(125) The bore **160** is sized and shaped to receive the driving tool (not shown) therethrough that engages the shank drive feature **46** when the shank body **6** is driven into bone with the receiver **10** attached. Also, the bore **160** may receive a manipulation tool used for releasing the alternative locking insert **14′** from a locked position with the receiver, the tool pressing down on the shank and also gripping the insert **14′** at the opposed through bores **166′** or with other tool engaging features. A manipulation tool for un-wedging the insert **14′** from the receiver **10** may also access the bores **166′** from the receiver through bores **74**. Referring back to the insert **14**, the outer notches defined by the surfaces **167** and **168** may also receive tools extending through receiver through bores **74** to temporarily lock the polyaxial mechanism as shown in FIGS. **46** and **48** and described in greater detail below. The illustrated insert **14** may further include other features, including additional grooves and recesses for manipulating and holding the insert **14** within the receiver **10** and providing adequate clearance between the retainer **12** and the insert **14**.

(126) With reference to FIGS. **1** and **30-33**, the illustrated elongate rod or longitudinal connecting member **21** (of which only a portion has been shown) can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface **22** of uniform diameter. The rod **21** may be made from a variety of metals, metal alloys, non-metals and deformable and less compressible plastics, including, but not limited to rods made of elastomeric, polyetheretherketone (PEEK) and other types of materials, such as polycarbonate urethanes (PCU) and polyethelenes.

(127) Longitudinal connecting members for use with the assembly **1** may take a variety of shapes, including but not limited to rods or bars of oval, rectangular or other curved or polygonal cross-section. The shape of the insert **14** may be modified so as to closely hold the particular longitudinal connecting member used in the assembly **1**. Some embodiments of the assembly **1** may also be used with a tensioned cord. Such a cord may be made from a variety of materials, including polyester or other plastic fibers, strands or threads, such as polyethylene-terephthalate. Furthermore, the longitudinal connector may be a component of a longer overall dynamic stabilization connecting member, with cylindrical or bar-shaped portions sized and shaped for being received by the compression insert **14** of the receiver having a U-shaped, rectangular- or

other-shaped channel, for closely receiving the longitudinal connecting member. The longitudinal connecting member may be integral or otherwise fixed to a bendable or damping component that is sized and shaped to be located between adjacent pairs of bone screw assemblies **1**, for example. A damping component or bumper may be attached to the longitudinal connecting member at one or both sides of the bone screw assembly **1**. A rod or bar (or rod or bar component) of a longitudinal connecting member may be made of a variety of materials ranging from deformable plastics to hard metals, depending upon the desired application. Thus, bars and rods of the invention may be made of materials including, but not limited to metal and metal alloys including but not limited to stainless steel, titanium, titanium alloys and cobalt chrome; or other suitable materials, including plastic polymers such as polyetheretherketone (PEEK), ultra-high-molecular weight-polyethylene (UHMWP), polyurethanes and composites, including composites containing carbon fiber, natural or synthetic elastomers such as polyisoprene (natural rubber), and synthetic polymers, copolymers, and thermoplastic elastomers, for example, polyurethane elastomers such as polycarbonate-urethane elastomers.

(128) With reference to FIGS. **1** and **30-33**, the closure structure or closure top **18** shown with the assembly **1** is rotatably received between the spaced arms **62** of the receiver **10**. It is noted that the closure **18** top could be a twist-in or slide-in closure structure. The illustrated closure structure **18** is substantially cylindrical and includes an outer helically wound guide and advancement structure **193** in the form of a flange that operably joins with the guide and advancement structure **72** disposed on the arms **62** of the receiver **10**. The flange form utilized in accordance with the present invention may take a variety of forms, including those described in Applicant's U.S. Pat. No. 6,726,689, which is incorporated herein by reference. Although it is foreseen that the closure structure guide and advancement structure could alternatively be a buttress thread, a square thread, a reverse angle thread or other thread like or non-thread like helically wound advancement structure, for operably guiding under rotation and advancing the closure structure **18** downward between the arms **62** and having such a nature as to resist splaying of the arms **62** when the closure structure **18** is advanced into the channel **64**, the flange form illustrated herein as described more fully in Applicant's U.S. Pat. No. 6,726,689 is preferred as the added strength provided by such flange form beneficially cooperates with and counters any reduction in strength caused by the any reduced profile of the receiver **10** that may more advantageously engage longitudinal connecting member components. The illustrated closure structure **18** also includes a top surface **194** with an internal drive **196** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive **196** is used for both rotatable engagement and, if needed, disengagement of the closure **18** from the receiver arms **62**. It is also foreseen that the closure structure **18** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **198** of the closure is planar and further includes a rim **199** for engagement and penetration into the surface **22** of the rod **21** in certain embodiments of the invention. It is noted that in some embodiments, the closure top bottom surface **198** may include a central point and in other embodiments need not include a point and/or the rim. The closure top **18** may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure **18** interior for a length of wire (not shown) inserted therein to provide a guide for insertion of the closure top into the receiver arms **62**.

(129) An alternative closure top **218** for use with a deformable rod, such as a PEEK rod **221**, is shown in FIGS. **41** and **42**. The top **218** is identical to the top **18** with the exception that a point or nub **289** is located on a domed surface **290** in lieu of the rim of the closure top **18**. The closure top

218 otherwise includes a guide and advancement structure **283**, a top **284**, an internal drive **286** and a bottom outer annular surface **288** that same or substantially similar to the respective guide and advancement structure **193**, top **194**, internal drive **196** and a bottom surface **198** described herein with respect to the closure top **18**. In some embodiments, the internal drive **286** is not as large as the drive **196** of the closure top **18**, such smaller drive providing for less force being placed on a deformable rod, for example, and not being required when a locking insert, for example, the insert **14′** discussed below is utilized in a bone screw assembly of the invention.

(130) Returning to the assembly **1**, preferably, the receiver **10**, the retainer **12** and the compression insert **14** are assembled at a factory setting that includes tooling for holding, alignment and manipulation of the component pieces. In some circumstances, the shank **4** is also assembled with the receiver **10**, the retainer **12** and the compression insert **14** at the factory. In other instances, it is desirable to first implant the shank **4**, followed by addition of the pre-assembled receiver, retainer and compression insert at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **4**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size and/or variety (e.g., surface treatment of roughening the upper portion **8** and/or hydroxyapatite on the shank **6**), with the receiver, retainer and compression insert. Allowing the surgeon to choose the appropriately sized or treated shank **4** advantageously reduces inventory requirements, thus reducing overall cost and improving logistics and distribution.

(131) Pre-assembly of the receiver **10**, retainer **12** and compression insert **14** is shown in FIGS. **21-23**. With particular reference to FIG. **21**, first the retainer **12** is inserted into the upper receiver opening **66**, leading with the cylindrical surface **130** with the top surface **122** and the bottom surface **124** facing opposed arms **62** of the receiver **10** (shown in phantom). The retainer **12** is then lowered in such sideways manner into the channel **64**, followed by tilting the retainer **12** at a location in the vicinity of the receiver apertures **74** such that the top surface **122** now faces the receiver opening **66** (also as shown in phantom). Thereafter, the retainer **12** is lowered into the receiver cavity **61** until the bottom surface **124** is seated on the annular lower cavity seating surface **104** as shown in solid lines in FIG. **21**.

(132) With reference to FIGS. **22** and **23**, the insert **14** is loaded into the receiver **10**. The insert **14** is loaded into the receiver through the opening **66** as shown in FIG. **22** with bottom surfaces **159** of the friction fit crown collets **158** facing the receiver cavity **61** and the surfaces **170** and **175** facing the planar arm surfaces **69** that define the channel **64** as the insert **14** is lowered into the channel **64**. When the insert **14** is lowered into the receiver, the side surfaces **170** are slidingly received by the opposed receiver inner arm surfaces **69** defining the channel **64** until the insert **14** reaches the stops **92**. The insert **14** may be pressed further downwardly until the insert **14** is captured within the receiver **10** as best shown in FIG. **23**, with the surfaces **170** being slightly pinched or pressed inwardly toward the receiver axis B to allow the opposed surfaces **170** to engage and then move downwardly past the receiver stops **92**, the stops **92** thereafter prohibiting upward movement of the insert **14** out of the receiver channel **64**. Specifically, if the insert **14** is moved upwardly toward the opening **66** of the receiver, the insert surfaces **173** abut against bottom surfaces **95** of the stops **92**, prohibiting further upward movement of the insert **14** unless a tool is used to pinch the surfaces **170** toward one another while moving the insert **14** upwardly toward the receiver opening **66**.

(133) With further reference to FIG. **23** and FIG. **24**, the insert **14** does not drop further downwardly toward the retainer **12** at this time because the outer arm surfaces **163** are engaged with and are slightly pressed inwardly by the receiver arm surfaces **82**. The insert **14** is thus held in the position shown in FIGS. **23** and **24** by the resilient arms **156** thereof resiliently pressing against the receiver inner cylindrical surfaces **82**, which is a desired position for shipping as an assembly along with the separate shank **4**. Thus, the receiver **10**, retainer **12** and insert **14** combination is now pre-assembled and ready for assembly with the shank **4** either at the factory, by surgery staff prior

to implantation, or directly upon an implanted shank **4** as will be described herein.
(134) As illustrated in FIG. **24**, the bone screw shank **4** or an entire assembly **1** made up of the assembled shank **4**, receiver **10**, retainer **12** and compression insert **14**, is screwed into a bone, such as the vertebra **17** (shown in phantom), by rotation of the shank **4** using a suitable driving tool (not shown) that operably drives and rotates the shank body **6** by engagement thereof at the internal drive **46**. Specifically, the vertebra **17** may be pre-drilled to minimize stressing the bone and have a guide wire (not shown) inserted therein to provide a guide for the placement and angle of the shank **4** with respect to the vertebra. A further tap hole may be made using a tap with the guide wire as a guide. Then, the bone screw shank **4** or the entire assembly **1** is threaded onto the guide wire utilizing the cannulation bore **50** by first threading the wire into the opening at the bottom **28** and then out of the top opening at the drive feature **46**. The shank **4** is then driven into the vertebra using the wire as a placement guide. It is foreseen that the shank and other bone screw assembly parts, the rod **21** (also having a central lumen in some embodiments) and the closure top **18** (also with a central bore) can be inserted in a percutaneous or minimally invasive surgical manner, utilizing guide wires and attachable tower tools mating with the receiver. When the shank **4** is driven into the vertebra **17** without the remainder of the assembly **1**, the shank **4** may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer.
(135) With further reference to FIG. **24**, the pre-assembled receiver, insert and retainer are placed above the shank upper portion **8** until the shank upper portion is received within the receiver opening **110**. With particular reference to FIGS. **25-29**, as the shank upper portion **8** is moved into the interior **61** of the receiver base, the shank upper portion **8** presses upwardly against the retainer **12**, moving the retainer **12** into the expansion portion of the receiver cavity **61** partially defined by the annular upper or ledge surface **91** and the cylindrical surface **99**. As the portion **8** continues to move upwardly toward the channel **64**, the surface **34** forces outward movement of the retainer **12** towards the cylindrical surface **99** as shown in FIG. **25**, with upward movement of the retainer **12** being prohibited by the annular surface **91**. With reference to FIGS. **26** and **27**, the retainer **12** begins to return to its neutral state as the center of the sphere of the head **8** (shown in dotted lines) passes beyond the center of the retainer expansion recess. With further reference to FIGS. **27-28**, at this time also, the spherical surface **34** moves into engagement with the discontinuous surface **181** of the insert **14** at the flex tabs or panels **184** of the crown collet extension portions **158**. The tabs **184** expand slightly outwardly to receive the surface **34** as best shown in FIG. **28**. With further reference to both FIG. **28** and FIG. **29**, the spherical surface **34** then enters into full frictional engagement with the panel inner surface **181**. At this time, the insert **14** panels and the surface **34** are in a fairly tight friction fit, the surface **34** being pivotable with respect to the insert **14** with some force. Thus, a tight, non-floppy ball and socket joint is now created between the insert **14** and the shank upper portion **8**, even before the retainer **12** drops down into a seated position on the receiver annular surface **104**.
(136) With reference to FIG. **29**, the receiver may then be pulled upwardly or the shank **4** and attached retainer **12** are then moved downwardly into a desired position with the retainer **12** seated on the surface **104**. Again, this may be accomplished by either an upward pull on the receiver **10** or, in some cases, by driving the shank **4** further into the vertebra **17**. At this time, the insert arms **156** spring outwardly into the receiver surface located beneath the ledge **84** and defined by the discontinuous cylindrical surface **86**, making it impossible to move the insert **14** upwardly without special tooling and making it impossible to move the retainer **12** out of the locking portion of the receiver chamber defined by the receiver seat **104** and the cylindrical surface **101**. In some embodiments, when the receiver **10** is pre-assembled with the shank **4**, the entire assembly **1** may be implanted at this time by inserting the driving tool (not shown) into the receiver and the shank drive **46** and rotating and driving the shank **4** into a desired location of the vertebra **17**. With reference to FIGS. **29** and **30** and also, for example, to FIGS. **46** and **48**, at this time, the receiver

10 may be articulated to a desired angular position with respect to the shank **4**, that will be held, but not locked, by the frictional engagement between the insert **14** flex tabs **184** and the shank upper portion **8**.

(137) With reference to FIGS. **30-33**, the rod **21** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **1**. The closure structure **18** is then advanced between the arms **62** of each of the receivers **10**. The closure structure **18** is rotated, using a tool engaged with the inner drive **196** until a selected pressure is reached at which point the rod **21** engages the U-shaped seating surface **153** of the compression insert **14**, pressing the stepped gripping surfaces **182** against the shank spherical surface **34**, the edges of the stepped surfaces **182** penetrating into the spherical surface **34**, pressing the shank upper portion **8** into locked frictional engagement with the retainer **12**. Specifically, as the closure structure **18** rotates and moves downwardly into the respective receiver **10**, the rim **199** engages and penetrates the rod surface **22**, the closure structure **18** pressing downwardly against and biasing the rod **21** into compressive engagement with the insert **14** that urges the shank upper portion **8** toward the retainer **12** inner body edge **128** and into locking engagement therewith, the retainer **12** bottom surface **124** frictionally abutting the surface **104** and the outer cylindrical surface **130** expanding outwardly and abutting against the receiver cylindrical surface **101** that defines the receiver locking chamber. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **6** with respect to the receiver **10**. If disassembly if the assembly **1** is desired, such is accomplished in reverse order to the procedure described previously herein for assembly.

(138) With reference to FIGS. **34-40**, an alternative lock-and-release compression insert **14′** is illustrated for use with the shank **4**, receiver **10**, retainer **12**, closure top **18** and rod **21** previously described herein, the resulting assembly identified as an assembly **1′** in FIGS. **39-40**, for example. The insert **14′** may be identical and is illustrated as substantially similar to the insert **14** previously described herein, with the exception that the insert **14′** is sized for a locking interference fit with the edges **97** and adjacent planar surfaces **71** of the receiver **10** as will be described in greater detail below. The illustrated insert **14′** also differs from the insert **14** in that the insert **14′** includes tool receiving apertures **166′** and does not have outwardly flared arms for resiliently engaging the receiver **10** during shipping and the early assembly steps.

(139) Thus, the locking insert **14** includes a body **150′**, a pair of opposed ends **151′**, a saddle surface **153′**, a pair of arms **156′**, a pair of arm upper surfaces **157′**, a pair of crown collet extensions **158′** having bottom surfaces **159′**, a bore **160′**, outer cylindrical arm surfaces **162′**, arm top surfaces **164′**, body bottom surfaces **165′**, a pair of v-shaped apertures that include outer sloping surfaces **167′**, and lower planar surfaces **168′**, extended portions with outer planar side surfaces **170′**, planar end surfaces **171′**, a pair of base extensions **172′**, upper surfaces **173′**, narrow lower ledges **174′**, inset planar side surfaces **175′**, planar end surfaces **177′**, bottom surfaces **178′**, an inner cylindrical surface **180′**, an inner spherical surface **181′** and an inner gripping surface portion **182′**, slits **183′** and flex panels **184′** with outer surfaces **185′** that are the same or substantially similar in form and function to the respective body **150**, pair of opposed ends **151**, saddle surface **153**, pair of arms **156**, pair of arm upper surfaces **157**, pair of crown collet extensions **158** having bottom surfaces **159**, bore **160**, outer cylindrical arm surfaces **162**, arm top surfaces **164**, body bottom surfaces **165**, a pair of v-shaped apertures that include outer sloping surfaces **167**, and lower planar surfaces **168**, extended portions with outer planar side surfaces **170**, planar end surfaces **171**, pair of base extensions **172**, upper surfaces **173**, narrow lower ledges **174**, inset planar side surfaces **175**, planar end surfaces **177**, bottom surfaces **178**, inner cylindrical surface **180**, inner spherical surface **181**, inner gripping surface portion **182**, slits **183** and flex panels **184** with outer surfaces **185** previously described herein with respect to the insert **14**. As mentioned above, the insert **14′** does not include flared upper outer surfaces **163**, but rather the cylindrical surfaces **162′** extend from the top surfaces **164′** to the body lower or bottom surfaces **165′**. Furthermore, the top arm surfaces **164′** are not sloping, but rather are planar surfaces disposed substantially parallel to the

bottom surfaces **165′**. The insert **14′** includes through holes **166′** for receiving manipulation tools, the holes **166′** formed on the arm surfaces **162′** and located directly above and adjacent to the sloping surfaces **167′**.
(140) The insert **14′** planar side surfaces **175′** are sized and shaped for a locking interference fit with the receiver at a lower portion of the receiver channel **64**. In other words, a width measured between surfaces **175′** is sized large enough to require that the insert **14′** must be forced into the space between the receiver surfaces **71** starting at the edge surfaces **97** by a tool or tools or by the closure top **18** forcing the rod **21** downwardly against the insert **14′** with sufficient force to interferingly lock the insert into the receiver between the planar surfaces **71**.
(141) With reference to FIGS. **37-40**, the insert **14′** is assembled with the receiver **10**, retainer **12**, shank **4**, rod **21** and closure top **18**, in a manner the same as previously described above with respect to the assembly **1**, resulting in an assembly **1′**, with the exception that the insert **14′** must be forced downwardly into a locking interference fit with the receiver **10** when the shank **4** is locked in place, as compared to the easily sliding relationship between the insert **14** and the receiver **10**. Also, the receiver arms **156′** do not engage the surfaces **82** of the receiver **10**. Rather, prior to assembly with the rod **21** and the closure top **18**, the compression insert **14′** outer surfaces **170′** are slidingly received by receiver surfaces **71**, but the surfaces **175′** are not. The insert **14′** is thus prohibited from moving any further downwardly at the edges **97** unless forced downwardly by a locking tool or by the closure top pressing downwardly on the rod that in turn presses downwardly on the insert **14′** as shown in FIGS. **38-40**. With further reference to FIG. **38**, at this time, the receiver **10** may be articulated to a desired angular position with respect to the shank **4**, such as that shown in FIGS. **46** and **48**, for example, that will be held, but not locked, by the frictional engagement between the insert panels **184′** and the shank upper portion **8**.
(142) The rod **21** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **1′**. The closure structure **18** is then inserted into and advanced between the arms **62** of each of the receivers **10**. The closure structure **18** is rotated, using a tool engaged with the inner drive **196** until a selected pressure is reached at which point the rod **21** engages the U-shaped seating surface **153′** of the compression insert **14′**, further pressing the insert stepped shank gripping surfaces **182′** against the shank spherical surface **34**, the edges of the stepped surfaces **182′** penetrating into the spherical surface **34**, pressing the shank upper portion **8** into locked frictional engagement with the retainer **12**. Specifically, as the closure structure **18** rotates and moves downwardly into the respective receiver **10**, the rim **199** engages and penetrates the rod surface **22**, the closure structure **18** pressing downwardly against and biasing the rod **21** into compressive engagement with the insert **14′** that urges the shank upper portion **8** toward the retainer **12** and into locking engagement therewith, the retainer **12** frictionally abutting the surface **104** and expanding outwardly against the cylindrical surface **101**. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **6** with respect to the receiver **10**. Tightening the helical flange form to 100 inch pounds can create 1000 pounds of force and it has been found that an interference fit is created between the planar surfaces **175′** of the insert **14′** and the edges **97** and planar surfaces **71** of the receiver at between about 700-900 inch pounds. So, as the closure structure **18** and the rod **21** press the insert **14′** downwardly toward the base of the receiver **10**, the insert surfaces **175′** are forced into the receiver at the edges **97**, thus forcing and fixing the insert **14** into frictional interference engagement with the receiver surfaces **71**.
(143) With reference to FIG. **41**, at this time, the closure top **18** may be loosened or removed and/or the rod **21** may be adjusted and/or removed and the frictional engagement between the insert **14′** and the receiver **10** at the insert surfaces **175′** will remain locked in place, advantageously maintaining a locked angular position of the shank **4** with respect to the receiver **10**.
(144) With further reference to FIGS. **41** and **42**, at this time, another rod, such as the deformable rod **221** and cooperating alternative closure top **218** may be loaded onto the already locked-up

assembly to result in an alternative assembly **201′**. As mentioned above, the closure drive **286** may advantageously be made smaller than the drive of the closure **18**, such that the deformable rod **221** is not unduly pressed or deformed during assembly since the polyaxial mechanism is already locked.

(145) With reference to FIGS. **43-45**, a two-piece tool **600** is illustrated for releasing the insert **14′** from the receiver **10**. The tool **600** includes an inner flexible tube-like structure with opposed inwardly facing prongs **612** located on either side of a through-channel **616**. The channel **616** may terminate at a location spaced from the prongs **612** or may extend further upwardly through the tool, resulting in a two-piece tool **610**. The tool **600** includes an outer, more rigid tubular member **620** having a smaller through channel **622**. The member **620** slidingly fits over the tube **610** after the flexible member **610** prongs **612** are flexed outwardly and then fitted over the receiver **10** and then Within through bores of the opposed apertures **74** of the receiver **10** and aligned opposed bores **166′** located on arms of the insert **14′**. In FIG. **43**, the tool **600** is shown during the process of unlocking the insert **14′** from the receiver **10** with the outer member **620** surrounding the inner member **610** and holding the prongs **612** within the receiver **10** and insert **14′** apertures while the tool **600** is pulled upwardly away from the shank **4**. It is foreseen that the tool **600** may further include structure for pressing down upon the receiver **10** while the prongs and tubular member are pulled upwardly, such structure may be located within the tool **600** and press down upon the top surfaces **73** of the receiver arms, for example.

(146) Alternatively, another manipulation tool (not shown) may be used that is inserted into the receiver at the opening **66** and into the insert channel formed by the saddle **153′**, with prongs or extensions thereof extending outwardly into the insert through bores **166′**; a piston-like portion of the tool thereafter pushing directly on the shank upper portion **8**, thereby pulling the insert **14′** away from the receiver surface **90** and thus releasing the polyaxial mechanism. At such time, the shank **4** may be articulated with respect to the receiver **10**, and the desired friction fit returns between the insert **14** flex panels and the shank surface **34**, so that an adjustable, but non-floppy relationship still exists between the shank **4** and the receiver **10**. If further disassembly if the assembly is desired, such is accomplished in reverse order to the procedure described previously herein for the assembly **1**.

(147) With reference to FIGS. **46-48**, another manipulation tool, generally **700** is illustrated for independently locking an insert **14′**, or as shown, for temporarily independently locking the non-locking insert **14** to the receiver **10**. The tool **700** includes a pair of opposed arms **712**, each having an engagement extension **716** positioned at an angle with respect to the respective arm **712** such that when the tool is moved downwardly toward the receiver, one or more inner surfaces **718** of the engagement extension **716** slide along the surfaces **77** of the receiver and surfaces **167** of the insert **14** to engage the insert **14**, with a surface **720** pressing downwardly on the insert surfaces **168** to lock the polyaxial mechanism of the assembly **1**. As shown in FIG. **48**, when the insert **14** is locked against the receiver **10**, the tool bottom surfaces **720** do not bottom out on the receiver surfaces **75′**, but remained spaced therefrom. In the illustrated embodiment, the surface **718** is slightly rounded and each arm extension **716** further includes a planar lower surface **722** that creates an edge with the bottom surface **720** for insertion and gripping of the insert **14** at the juncture of the surface **167** and the surface **168**. The tool **700** may include a variety of holding and pushing/pulling mechanisms, such as a pistol grip tool, that may include a ratchet feature, a hinged tool, or, a rotatably threaded device, for example.

(148) With reference to FIGS. **49-83** the reference number **1001** generally represents another polyaxial bone screw apparatus or assembly according to the present invention. The assembly **1001** includes a shank **1004**, that further includes a body **1006** integral with an upwardly extending upper portion or head-like capture structure **1008**; a receiver **1010**; a retainer structure illustrated as a resilient open ring **1012**, and a friction fit crown collet compression or pressure insert **1014**. The receiver **1010**, retainer **1012** and compression insert **1014** are initially assembled and may be

further assembled with the shank **1004** either prior or subsequent to implantation of the shank body **1006** into a vertebra **1017**, as will be described in greater detail below. FIGS. **49** and **82-83** further show a closure structure **1018** for capturing a longitudinal connecting member, for example, a rod **1021** which in turn engages the compression insert **1014** that presses against the shank upper portion **1008** into fixed frictional contact with the retainer **1012**, so as to capture, and fix the longitudinal connecting member **1021** within the receiver **1010** and thus fix the member **1021** relative to the vertebra **1017**. The illustrated rod **1021** is substantially similar in form and function to the rod **21** previously described herein. It is foreseen that in other embodiments, the rod **1021** may be elastic, deformable and/or of a different cross-sectional geometry. The receiver **1010** and the shank **1004** cooperate in such a manner that the receiver **1010** and the shank **1004** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **1010** with the shank **1004** until both are locked or fixed relative to each other near the end of an implantation procedure.

(149) The shank **1004**, best illustrated in FIG. **49-51**, is elongate, with the shank body **1006** having a helically wound bone implantable thread **1024** (single or dual lead thread form) extending from near a neck **1026** located adjacent to the upper portion or head **1008**, to a tip **1028** of the body **1006** and extending radially outwardly therefrom. During use, the body **1006** utilizing the thread **1024** for gripping and advancement is implanted into the vertebra **1017** leading with the tip **1028** and driven down into the vertebra with an installation or driving tool (not shown), so as to be implanted in the vertebra to a location at or near the neck **1026**. The shank **1004** has an elongate axis of rotation generally identified by the reference letter A.

(150) The neck **1026** extends axially upward from the shank body **1006**. The neck **1026** may be of the same or is typically of a slightly reduced radius as compared to an adjacent upper end or top **1032** of the body **1006** where the thread **1024** terminates. Further extending axially and outwardly from the neck **1026** is the shank upper portion or head **1008** that provides a connective or capture apparatus disposed at a distance from the upper end **1032** and thus at a distance from the vertebra **1017** when the body **1006** is implanted in such vertebra.

(151) The shank upper portion **1008** is configured for a pivotable connection between the shank **1004** and the retainer **1012** and receiver **1010** prior to fixing of the shank **1004** in a desired position with respect to the receiver **1010**. The shank upper portion **1008** has an outer, convex and substantially spherical surface **1034** that extends outwardly and upwardly from the neck **1026** and terminates at a substantially planar top or rim surface **1038**. The spherical surface **1034** has an outer radius configured for frictional, non-floppy, sliding cooperation with a discontinuous concave surface of the compression insert **1014**, as well as ultimate frictional engagement and penetration by a stepped, gripping portion of the insert **1014**. The top surface **1038** is substantially perpendicular to the axis A. The spherical surface **1034** shown in the present embodiment is substantially smooth, but in some embodiments may include a roughening or other surface treatment and is sized and shaped for cooperation and ultimate frictional engagement with the compression insert **1014** as well as ultimate frictional engagement with the retainer **1012**. The shank spherical surface **1034** is locked into place exclusively by the insert **1014** and the retainer **1012** and not by inner surfaces defining the receiver cavity.

(152) A counter sunk substantially planar base or stepped seating surface **1045** partially defines an internal drive feature or imprint **1046**. The illustrated internal drive feature **1046** is an aperture formed in the top surface **1038** and has a hex shape designed to receive a hex tool (not shown) of an Allen wrench type, into the aperture for rotating and driving the bone screw shank **1004**. It is foreseen that such an internal tool engagement structure may take a variety of tool-engaging forms and may include one or more apertures of various shapes, such as a pair of spaced apart apertures or a multi-lobular or star-shaped aperture, such as those sold under the trademark TORX, or the like. The seat or base surfaces **1045** of the drive feature **1046** are disposed substantially

perpendicular to the axis A with the drive feature **1046** otherwise being coaxial with the axis A. As illustrated in FIGS. **50** and **51**, the drive seat **1045** may include beveled or stepped surfaces that may further enhance gripping with the driving tool. In operation, a driving tool (not shown) is received in the internal drive feature **1046**, being seated at the base **1045** and engaging the six faces of the drive feature **1046** for both driving and rotating the shank body **1006** into the vertebra **1017**, either before the shank **1004** is attached to the receiver **1010** or after the shank **1004** is attached to the receiver **1010**, with the shank body **1006** being driven into the vertebra **1017** with the driving tool extending into the receiver **1010**.

(153) The shank **1004** shown in the drawings is cannulated, having a small central bore **1050** extending an entire length of the shank **1004** along the axis A. The bore **1050** is defined by an inner cylindrical wall of the shank **1004** and has a circular opening at the shank tip **1028** and an upper opening communicating with the external drive **1046** at the driving seat **1045**. The bore **1050** is coaxial with the threaded body **1006** and the upper portion **1008**. The bore **1050** provides a passage through the shank **1004** interior for a length of wire (not shown) inserted into the vertebra **1017** prior to the insertion of the shank body **1006**, the wire providing a guide for insertion of the shank body **1006** into the vertebra **1017**. The shank body **1006** may be treated or coated as previously described herein with respect to the shank body **6** of the assembly **1**.

(154) With particular reference to FIGS. **49** and **52-57**, the receiver **1010** has a generally U-shaped appearance with partially discontinuous and partially cylindrical inner and outer profiles. The receiver **1010** has an axis of rotation B that is shown in FIG. **49** as being aligned with and the same as the axis of rotation A of the shank **1004**, such orientation being desirable, but not required during assembly of the receiver **1010** with the shank **1004** (see, e.g., FIG. **76** showing the receiver **1010** being “popped on” to a shank **1004** that is implanted in a vertebra **1017** and disposed at an angle with respect to the receiver). After the receiver **1010** is pivotally attached to the shank **1004**, either before or after the shank **1004** is implanted in a vertebra **1017**, the axis B is typically disposed at an angle with respect to the axis A, as shown, for example, in FIG. **96**.

(155) The receiver **1010** includes a substantially cylindrical base **1060** defining a bore or inner cavity, generally **1061**, the base **1060** being integral with a pair of opposed upstanding arms **1062** forming a cradle and defining a channel **1064** between the arms **1062** with an upper opening, generally **1066**, and a U-shaped lower channel portion or seat **1068**, the channel **1064** having a width for operably snugly receiving the rod **1021** or portion of another longitudinal connector, such as a sleeve of a tensioned cord connector or other soft or dynamic connector between the arms **1062**, the channel **1064** communicating with the base cavity **1061**. Outer front and rear opposed substantially planar arm surfaces **1069** partially define the channel **1064** directly above the seat **1068**, the surfaces **1069** advantageously reduce the run on the rod (i.e., provide a more narrow receiver portion that in turn provides more space and thus more access between bone anchors along the rod or other connecting member) and provide the planar surface **1069** for flush or close contact with other connecting member components in certain embodiments, such as sleeves, bumpers or spacers that cooperate with rods or cord-type connecting members.

(156) Each of the arms **1062** has an interior surface, generally **1070**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **1072** located adjacent top surfaces **1073** of each of the arms **1062**. In the illustrated embodiment, the guide and advancement structure **1072** is a partial helically wound interlocking flange form configured to mate under rotation with a similar structure on the closure structure **1018**, as described more fully below. However, it is foreseen that for certain embodiments of the invention, the guide and advancement structure **1072** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-thread-like helically wound discontinuous advancement structures, for operably guiding under rotation and advancing the closure structure **1018** downward between the arms **1062**, as well as eventual torquing when the closure structure **1018** abuts against the rod **1021** or other longitudinal connecting member. It is

foreseen that the arms could have break-off extensions.

(157) An opposed pair of upper rounded off triangular or delta-shaped tool receiving and engaging apertures, generally **1074**, each having a through bore formed by an upper arched surface **1075** and a substantially planar bottom surface **1075′**, are formed on outer surfaces **1076** of the arms **1062**. Each through bore surface **1075** and **1075′** extends through the arm inner surface **1070**. The apertures **1074** with through bore portions **1075** and **1075′** are sized and shaped for receiving locking, unlocking and other manipulation tools and may aid in receiving and downloading the retainer ring **1012** during top loading of the retainer **1012** into the receiver **1010**. Each aperture **1074** further includes a sloping tool alignment surface **1077** that surrounds the arched bore portion **1075** and does not extend completely through the respective arm **1062**. In the present embodiment, part of the receiver defining the surface **1077** located along the arched aperture portion is crimped into the insert **1014** during assembly thereof with the receiver **1010** as will be described in greater detail below. In other embodiments of the invention, other walls or surfaces defining the aperture **1074** or other material defining other apertures or grooves may be inwardly crimped. It is noted that the illustrated receiver **1010** is an integral structure and devoid of any spring tabs or collet-like structures. Alternatively, in some embodiments, spring tabs or other movable structure may be included on the receiver **1010** or the insert **1014** for retaining the insert **1014** in a desired position, with regard to rotation and axial movement (along the axis A) with respect to the receiver **1010**. Preferably the insert and/or receiver are configured with structure for blocking rotation of the insert with respect to the receiver, but allowing some up and down movement of the insert with respect to the receiver during the assembly and implant procedure.

(158) Located directly centrally above each delta shaped aperture **1074** is a cylindrical depression or aperture **1078**, also formed in each arm surface **1076**, but not extending through the inner surface **1070**. Two additional pair of tool receiving and engaging apertures **1079** are also formed in the front and rear surfaces **1069** of the receiver arms **1062**. Transition base surfaces **1080** span between the planar surfaces **1069** at the U-shaped seat **1068** and the cylindrical base **1060**, the surfaces **1080** sloping downwardly toward the base **1060** at an angle with respect to the axis B. Some or all of the apertures **1074**, **1078** and **1079** may be used for holding the receiver **1010** during assembly with the insert **1014**, the retainer **1012** and the shank **1004**; during the implantation of the shank body **1006** into a vertebra when the shank is pre-assembled with the receiver **1010**; during assembly of the bone anchor assembly **1001** with the rod **1021** and the closure structure **1018**; and during lock and release adjustment of the some inserts of the invention with respect to the receiver **1010**, either into or out of frictional engagement with the inner surfaces of the receiver **1010** as will be described in greater detail below. It is foreseen that tool receiving grooves, depressions or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **1062**.

(159) Returning to the interior surface **1070** of the receiver arms **1062**, located below the guide and advancement structure **1072** is a discontinuous cylindrical surface **1082** partially defining a run-out feature for the guide and advancement structure **1072**. The cylindrical surface **1082** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure **1072**. Moving downwardly, in a direction toward the base **1060**, adjacent the cylindrical surface **1082** of each arm is a run-out seat or surface **1084** that extends inwardly toward the axis B and slopes toward the axis B. Adjacent to and located below the surface **1084** is another cylindrical surface **1086** having a diameter smaller than the diameter of the surface **1082**. The through bore surfaces **1075** and **1075′** extend through the arms primarily at the surfaces **1086**, with an upper portion of each arch **1075** extending through one of the surfaces **1082**. Located near each aperture surface **1075** is an inner surface portion **1087** that engages the insert **1014** when the thin wall at the surface **1077** is crimped toward the insert **1014** during assembly of such insert in the receiver **1010** as will be described in greater detail below. A continuous annular surface **1088** is located below and adjacent to the cylindrical surface **1086**. The otherwise discontinuous cylindrical surface **1086** that

partially forms the receiver arms **1062** is also continuous at and near the interface between the surface **1086** and the annular surface **1088**, forming an upper portion of the receiver cavity **1061**. The surface **1088** is disposed substantially perpendicular or slopes inwardly toward the axis B. A continuous inner cylindrical surface **1090** is adjacent the annular surface **1088** and extends downwardly into the receiver base **1060**. The surface **1090** is disposed parallel to the receiver axis B. The surface **1090** has a diameter slightly smaller than the diameter of the surface **1086**. The cylindrical surfaces **1086** and **1090** are sized to receive portions of the insert **1014**, and as will be described in greater detail below, in some embodiments provide a locking interference fit with a cylindrical portion of a locking insert.

(160) Now, with respect to the base **1060** and more specifically, the base cavity **1061**, a lower portion of the surface **1090** that extends into the base and partially defines the base cavity **1061** terminates at an annular surface or ledge **1095**. The ledge **1095** extends outwardly away from the axis B and is substantially perpendicular thereto.

(161) Extending downwardly from the ledge **1095** is a cylindrical surface **1099** that partially defines the base cavity **1061** and in particular, defines an expansion chamber for the retainer **1012**. The cylindrical surface **1099** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded retainer **1012**. The surfaces **1095** and **1099** define a circumferential recess that is sized and shaped to receive the retainer **1012** as it expands around the shank upper portion **1008** as the shank **1004** moves upwardly toward the channel **1064** during assembly, as well as form a stop or restriction to prevent the expanded retainer **1012** from moving upwardly with the shank portion **1008**, the surface **1095** preventing the retainer **1012** from passing upwardly out of the cavity **1061** whether the retainer **1012** is in a partially or fully expanded position or state. A cylindrical surface **1101** located below the cylindrical surface **1099** is sized and shaped to closely receive the retainer **1012** when the retainer is in a neutral position as shown in FIG. **71**, for example. Thus, the cylindrical surface **1101** has a diameter smaller than the diameter of the cylindrical surface **1099** that defines the expansion area for the retainer **1012**. The surface **1101** is joined or connected to the surface **1099** by one or more beveled, curved or conical surfaces **1102**. The surfaces **1102** allow for sliding gradual movement of the retainer **1012** into the space or locking chamber defined by the surface **1101** and ultimate seating of the retainer **1012** on a lower annular surface **1104** located below and adjacent to the cylindrical surface **1101**.

(162) Located below and adjacent to the annular seating surface **1104** is another substantially cylindrical surface **1106** that communicates with a beveled or flared bottom opening surface **1107**, the surface **1107** communicating with an exterior base surface **1108** of the base **1060**, defining a lower opening, generally **1110**, into the base cavity **1061** of the receiver **1010**.

(163) With particular reference to FIGS. **49** and **58-62**, the lower open or split retainer **1012**, that operates to capture the shank upper portion **1008** within the receiver **1010**, has a central axis that is operationally the same as the axis B associated with the receiver **1010** when the shank upper portion **1008** and the retainer **1012** are installed within the receiver **1010**. The retainer ring **1012** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **1012** may be expanded during various steps of assembly as will be described in greater detail below. The retainer **1012** has a central channel or hollow through bore, generally **1121**, that passes entirely through the ring **1012** from a top surface **1122** to a bottom surface **1124** thereof. Surfaces that define the channel or bore **1121** include a discontinuous inner cylindrical surface **1125** adjacent the top surface **1122** and a discontinuous frusto-conical surface **1127** adjacent the surface **1125**, both surfaces coaxial when the retainer **1012** is in a neutral, non-expanded orientation. An edge **1128** is defined by the juncture of the top surface **1122** and the cylindrical surface **1125**. The edge **1128** may include a chamfer or bevel. As shown, for example, in FIG. **82**, the shank upper portion **1008** ultimately frictionally engages the retainer **1012** at the edge **1128** when the assembly **1001** is locked into a final position. The retainer **1012** further includes an outer cylindrical surface **1130** located adjacent the top surface **1122** and an outer beveled or frusto-conical surface **1132** adjacent

the bottom surface **1124**. The surface **1130** is oriented parallel to the central axis of the retainer **1012**. In some embodiments of the invention, spaced notches (not shown) may be formed in the cylindrical surface **1130** to receive a holding and manipulation tool (not shown). In some embodiments further notches on inner or outer surfaces of the retainer may be made to evenly distribute stress across the entire retainer **1012** during expansion thereof.

(164) The resilient retainer **1012** further includes first and second end surfaces, **1134** and **1135** disposed in spaced relation to one another when the retainer is in a neutral non-compressed state. The surface **1134** and **1135** may also be touching when the retainer is in a neutral state. Both end surfaces **1134** and **1135** are disposed substantially perpendicular to the top surface **1122** and the bottom surface **1124**. A width X between the surfaces **1134** and **1135** is very narrow (slit may be made by EDM process) to provide stability to the retainer **1012** during operation. Because the retainer **1012** is top loadable in a neutral state and the retainer **1012** does not need to be compressed to fit within the receiver cavity **1061**, the width X may be much smaller than might be required for a bottom loaded compressible retainer ring. The gap X functions only in expansion to allow the retainer **1012** to expand about the shank upper portion **1008**. This results in a stronger retainer that provides more surface contact with the shank upper portion **1008** upon locking, resulting in a sturdier connection with less likelihood of failure than a retainer ring having a greater gap. Furthermore, because the retainer **1012** is only expanded and never compressed inwardly, the retainer **1012** does not undergo the mechanical stress that typically is placed on spring ring type retainers known in the prior art that are both compressed inwardly and expanded outwardly during assembly.

(165) It is foreseen that in some embodiments of the invention, the retainer **1012** inner surfaces may include a roughening or additional material to increase the friction fit against the shank upper portion **1008** prior to lock down by the rod **1021** or other longitudinal connecting member. Also, the embodiment shown in FIGS. **58-62** illustrates the surfaces **1134** and **1135** as substantially parallel to the central axis of the retainer, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle.

(166) With particular reference to FIGS. **49** and **63-70**, the friction fit, crown compression insert **1014** is illustrated that is sized and shaped to be received by and down-loaded into the receiver **1010** at the upper opening **1066**. The compression insert **1014** has an operational central axis that is the same as the central axis B of the receiver **1010**. In operation, the insert advantageously frictionally engages the bone screw shank upper portion **1008**, allowing for un-locked but non-floppy placement of the angle of the shank **1004** with respect to the receiver **1010** during surgery prior to locking of the shank with respect to the receiver near the end of the procedure. As will be described in greater detail below with respect to the insert **1014′** illustrated in FIGS. **84-92**, in some embodiments of the invention, the insert that has locked the shank **1004** in a desired angular position with respect to the receiver **1010**, by, for example, compression from the rod **1021** and closure top **1018**, is also forced into an interference fit engagement with the receiver **1010** at the inner cylindrical surface **1090** and thus is capable of retaining the shank **1006** in a locked position even if the rod **1021** and closure top **1018** are removed. Such locked position may also be released by the surgeon if desired. The non-locking insert **1014** as well as the locking insert **1014′** are preferably made from a solid resilient material, such as a stainless steel or titanium alloy, so that portions of the insert may be snapped or popped onto the shank upper portion **1008** as well as pinched or pressed against and un-wedged from the receiver **1010** with a release tool.

(167) The non-locking crown collet compression insert **1014** includes a substantially cylindrical body **1136** integral with a pair of upstanding arms **1137** at an upper end thereof and integral with downwardly extending resilient superstructure in the form of an opposed pair of crown Collet extensions **1138** at a lower end thereof. A bore, generally **1140**, is disposed primarily within and through the body **1136** and communicates with a generally U-shaped through channel formed by a saddle **1141** that is partially defined by the upstanding arms **1137** and partially by the body **1136**.

The saddle **1141** is sized and shaped to closely, snugly engage the cylindrical rod **1021** and includes a curved lower seat **1142**. It is foreseen that an alternative embodiment may be configured to include planar holding surfaces that closely hold a square or rectangular bar as well as hold a cylindrical rod-shaped, cord, or sleeved cord longitudinal connecting member. The arms **1137** disposed on either side of the saddle **1141** extend upwardly from the body **1136**. The arms **1137** are sized and configured for ultimate placement at or near the cylindrical run-out surface **1082** and inner surfaces **1084** and **1086** located below the receiver guide and advancement structure **1072**. It is foreseen that in some embodiments of the invention, the insert arms **1137** may be extended and the closure top configured such the arms ultimately directly engage the closure top **1018** for locking of the polyaxial mechanism, for example, when the rod **1021** is made from a deformable material. In such embodiments, the insert **1014** would include a rotation blocking structure or feature on an outer surface thereof that abuts against cooperating structure located on an inner wall of the receiver **1010**, preventing rotation of the insert with respect to the receiver when the closure top is rotated into engagement with the insert. In the present embodiment, the arms **137** include outer surfaces **1143** that are illustrated as partially cylindrical and run from substantially planar top surfaces **1144** to near bottom surfaces **1148** of the collet extensions **1138**, the top surfaces **1144** ultimately being positioned in spaced relation with the closure top **1018**, so that the closure top **1018** frictionally engages the rod **1021** only, pressing the rod **1021** downwardly against the seating surface **1142**, the insert **1014** in turn pressing against the shank **1004** upper portion **1008** that presses against the retainer **1012** to lock the polyaxial mechanism of the bone screw assembly **1001** at a desired angle. Specifically, the illustrated partially cylindrical surfaces **1143** each extend from one of the arm top surfaces **1144** to a rim or ledge **1150** spaced from the bottom surfaces **1148** of the collet extensions **1138**. Each rim or ledge **1150** is positioned near an annular lower surface **1151** of the insert body **1136**. Located between each rim **1150** and each extension bottom surface **1148** is a discontinuous outer cylindrical surface **1152** partially defining each collet extension **1138** and having an outer diameter smaller than a diameter of the arm cylindrical surfaces **1143**. A through slot **1154** is centrally formed in each collet extension **1138** extending through the extension at the bottom surface **1148**, the outer cylindrical surface **1152** and partially into the outer cylindrical surface **1143** at the insert body **1136**. The illustrated embodiment includes the one slot **1154** centrally located in each extension **1138**. It is foreseen that other embodiments of the invention may include more or fewer slots **1154**. The slots **1154** substantially equally partition each of the extensions **1138**, forming four distinct resilient, fingers, tabs or panels **1154** that extend to the bottom surface **1148**.

(168) The surfaces **1143** are sized and shaped to generally fit within the receiver arms **1062**. The arm outer surfaces **1143** further include notches or grooves formed thereon for receiving manipulation, unlocking and locking tools. Although not shown, each surface **1143** may include one or more through bores or other apertures for receiving tooling, such as that shown on the locking insert **1014′** in FIG. **84**, for example. Centrally located (in some embodiments below a through bore) and formed in each surface **1143** is a delta or triangular notch or recess, generally **1156**, for receiving tooling defined in part by an upper sloping surface **1157** and intersecting a lower planar surface **1158** disposed substantially perpendicular to a central axis of the insert **1014** (and the axis B of the receiver when the insert is disposed within the receiver). Each of the surfaces **1167** and surface **1168** cooperate and align with the respective receiver aperture through bore surfaces **1077** and **1075′** when the insert **1014** is captured and operationally positioned within the receiver **1010** as will be described in greater detail below. In the illustrated embodiments, also formed in each surface **1143** are a pair of spaced v- or squared-off notches or grooves **1160** and **1161** that run from the respective top surface **1144** to near the surface **1158** of the central delta cut or notch **1156**. The grooves **1160** and **1161** cooperate with the receiver crimp wall inner surfaces **1087** to aid in alignment of the insert channel saddle **1141** with the receiver channel **1064** as shown, for example in FIGS. **74** and **75**. The illustrated pair of grooves **1160** and **1161** slope

slightly toward one another as best shown in FIG. **66**, running from respective bottom surfaces **1162** and **1163** to a closest location at the arm top surface **1144**.
(169) The u-shaped channel formed by the saddle **1141** is also partially defined by inner planar surfaces **1165** located near the arm top surfaces **1144**. The saddle **1141** also communicates with the bore **1140** at an inner cylindrical surface **1166**, the surface **1166** located centrally within the insert body **1136** and further communicating with a lower collet space that extends to the discontinuous bottom surfaces **1148** of the collet extensions **1138**. The inner cylindrical surface **1166** also communicates with a lower concave surface portion **1168** having a generally spherical profile with a radius the same or substantially similar to a radius of the surface **1034** of the shank upper portion or head **1008**. The surface **1168** primarily terminates at the body lower surface **1151**, but also extends into and partially defines inner surfaces of the collet extensions **1138**. Located along the inner surface **1168** between the cylindrical surface **1166** and the body lower surface **1151** is a shank gripping surface portion **1170**. The gripping surface portion **1170** includes one or more stepped surfaces or ridges sized and shaped to grip and penetrate into the shank head **1008** when the insert **1014** is locked against the head surface **1034**. In the illustrated embodiments, substantially all the inner surface **1168** is made up of stepped portions and thus the entire surface may also be described as having a plurality of cylindrical surfaces of graduating diameters. It is foreseen that the stepped surface portion **1170** may include a greater or fewer number of stepped surfaces and cover greater or less surface area of the inner surface **1168** having the same or similar spherical profile as the surface **1034** of the shank head **1008**. It is foreseen that any and all of the shank gripping surface portion **1170** and the surface **1168** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the shank upper portion **1008**. Located below the body lower surface **1151**, each collet extension **1138** includes a pair of inner upper curved portion **1172** and a pair of planar surface portion **1173**, each planar surface portion **1173** located between an adjacent curved portion **1172** and adjacent respective bottom surface **1148**. The planar surface portion **1173** are sized, shaped and positioned with respect to one another to provide a non-locking frictional fit with the spherical surface **1034** of the shank upper portion or head **1008** as will be described in greater detail below. It is foreseen that in some embodiments of the invention, the planar surfaces **1173** may be replaced by radiused surfaces having the same, greater or lesser radius than the shank surface **1034**.
(170) The insert bore **1140** is sized and shaped to receive the driving tool (not shown) therethrough that engages the shank drive feature **46** when the shank body **1006** is driven into bone with the receiver **1100** attached. Also, the bore **1140** may receive a manipulation tool used for releasing the alternative locking insert **1014′** from a locked position with the receiver, the tool pressing down on the shank and also gripping the insert **1014′** at the opposed through bores or with other tool engaging features. A manipulation tool for un-wedging the insert **1104′** from the receiver **1100** may also access the such tooling bores from the receiver through bores **1074**. The illustrated insert **1014** may further include other features, including grooves and recesses for manipulating and holding the insert **1014** within the receiver **1010** and providing adequate clearance between the retainer **1012** and the insert **1014**.
(171) The insert body **1136** located between the arms **1137** and the collet extensions **1138** has an outer diameter slightly smaller than a diameter between crests of the guide and advancement structure **1072** of the receiver **1010**, allowing for top loading of the compression insert **1014** into the receiver opening **1066**, with the arms **1137** of the insert **1014** being located between the receiver arms **6102** during insertion of the insert **1014** into the receiver **1010**. Once the arms **1137** of the insert **1014** are generally located beneath the guide and advancement structure **1072**, the insert **1014** is rotated into place about the receiver axis B until the top surfaces **1144** are located directly below the guide and advancement structure **1072** as will be described in greater detail below.

(172) With reference to FIGS. **49** and **82-83**, the illustrated elongate rod or longitudinal connecting member **1021** (of which only a portion has been shown) can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface **1022** of uniform diameter. The illustrated rod **1021** is identical or substantially similar to the rod **21** previously described herein. Other longitudinal connecting members for use with the assembly **1001** may take a variety of shapes, including but not limited to rods or bars of oval, rectangular or other curved or polygonal cross-section. The shape of the insert **1014** may be modified so as to closely hold, and if desired, fix or slidingly capture the longitudinal connecting member to the assembly **1001**. Some embodiments of the assembly **1001** may also be used with a tensioned cord. Such a cord may be made from a variety of materials, including polyester or other plastic fibers, strands or threads, such as polyethylene-terephthalate. Furthermore, the longitudinal connector may be a component of a longer overall dynamic stabilization connecting member, with cylindrical or bar-shaped portions sized and shaped for being received by the compression insert **1014** of the receiver having a U-shaped, rectangular- or other-shaped channel, for closely receiving the longitudinal connecting member. The longitudinal connecting member may be integral or otherwise fixed to a bendable or damping component that is sized and shaped to be located between adjacent pairs of bone screw assemblies **1001**, for example. A damping component or bumper may be attached to the longitudinal connecting member at one or both sides of the bone screw assembly **1001**. A rod or bar (or rod or bar component) of a longitudinal connecting member may be made of a variety of materials ranging from non-metallic materials, such as soft deformable plastics, to hard metals, depending upon the desired application. Thus, bars and rods of the invention may be made of materials including, but not limited to metal and metal alloys including but not limited to stainless steel, titanium, titanium alloys and cobalt chrome; or other suitable materials, including plastic polymers such as polyetheretherketone (PEEK), ultra-high-molecular weight-polyethylene (UHMWP), polyurethanes and composites, including composites containing carbon fiber, natural or synthetic elastomers such as polyisoprene (natural rubber), and synthetic polymers, copolymers, and thermoplastic elastomers, for example, polyurethane elastomers such as polycarbonate-urethane elastomers.

(173) With reference to FIGS. **49** and **82-83**, the closure structure or closure top **1018** shown with the assembly **1001** is identical or substantially similar to the closure top **18** previously described herein with respect to the assembly **1**. It is noted that the closure **1018** top could be a twist-in or slide-in closure structure. The illustrated closure structure **1018** is substantially cylindrical and includes a an outer helically wound guide and advancement structure **1182** in the form of a flange that operably joins with the guide and advancement structure **1072** disposed on the arms **1062** of the receiver **1010**. The illustrated closure structure **1018** also includes a top surface **1184** with an internal drive **1186** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive **1166** is used for both rotatable engagement and, if needed, disengagement of the closure **1018** from the receiver arms **1062**. It is also foreseen that the closure structure **1018** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **1188** of the closure is planar and further includes a rim **1190** and may or may not include a further include a central point (not shown), the rim **1190** and or the point (not shown) for engagement and penetration into the surface **1022** of the rod **1021** in certain embodiments of the invention. The closure top **1018** may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure

1018 interior for a length of wire (not shown) inserted therein to provide a guide for insertion of the closure top into the receiver arms **1062**.

(174) An alternative closure top **1218** for use with a deformable rod, such as a PEEK rod **1221**, is shown in FIGS. **91** and **92**. The top **2118** is identical to the top **1018** with the exception that a point or nub **1289** is located on a domed surface **1290** in lieu of the rim of the closure top **1018**. The closure top **1218** otherwise includes a guide and advancement structure **1283**, a top **1284**, an internal drive **1286** and a bottom outer annular surface **1288** that are the same or substantially similar to the guide and advancement structure **1183**, top **1184**, internal drive **1186** and a bottom surface **1188** described herein with respect to the closure top **1018**. In some embodiments, the internal drive **1286** is not as large as the drive **1186** of the closure top **1018**, such smaller drive providing for less force being placed on a deformable rod, for example, and not being required when a locking insert, for example, the insert **1014′** discussed below is utilized in a bone screw assembly of the invention.

(175) Returning to the assembly **1110**, preferably, the receiver **1010**, the retainer **1012** and the compression insert **1014** are assembled at a factory setting that includes tooling for holding, pressing and alignment of the component pieces as well as compressing or expanding the insert **1014** arms and/or collet extensions, if needed, as well as crimping a portion of the receiver **1010** toward the insert **1014**. In some circumstances, the shank **1004** is also assembled with the receiver **1010**, the retainer **1012** and the compression insert **1014** at the factory. In other instances, it is desirable to first implant the shank **1004**, followed by addition of the pre-assembled receiver, retainer and compression insert at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **1004**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size and/or variety (e.g., surface treatment of roughening the upper portion **1008** and/or hydroxyapatite on the shank **1006**), with the receiver, retainer and compression insert. Allowing the surgeon to choose the appropriately sized or treated shank **1004** advantageously reduces inventory requirements, thus reducing overall cost.

(176) Pre-assembly of the receiver **1010**, retainer **1012** and compression insert **1014** is shown in FIGS. **71-75**. First, the retainer **1012** is downloaded in a sideways manner into the receiver **1010** through the upper opening **1066** with the outer surface **1130** facing the receiver channel seat **1068**. The retainer **1012** is then lowered between the arms **1062** and toward the receiver base **1060** as shown in phantom in FIG. **71**, the retainer being turned or tilted to a position within the receiver base **1060** inner cavity **1061** wherein the retainer bottom surface **1124** is manipulated to a position facing the annular surface **1104** (also shown in phantom) and then fully seated upon the inner base annular surface **1104** as shown in solid lines in FIG. **71**. With reference to FIG. **72**, the compression insert **1014** is then downloaded into the receiver **1010** through the upper opening **1066** with the crown collet extension bottom surfaces **1148** facing the receiver arm top surfaces **1073** and the insert arms **1137** as well as the insert collet extensions **1138** located between the opposed receiver arms **1062**. The insert **1014** is then lowered toward the channel seat **1068** until the insert **1014** arm upper surfaces **1144** are adjacent the run-out area defined by the surfaces **1082** of the receiver below the guide and advancement structure **1072**. Thereafter, the insert **1014** is rotated in a clockwise or counter-clockwise manner about the receiver axis B until the upper arm surfaces **1144** are directly below the guide and advancement structure **1072** as illustrated in FIG. **73** with the U-shaped channel **1141** of the insert **1014** aligned with the U-shaped channel **1064** of the receiver **1010**. In some embodiments, the insert arms **1137** and collet extensions **1138** may need to be compressed slightly during rotation to clear inner surfaces of the receiver arms **1062**. As shown in FIGS. **73-75**, the outer cylindrical surfaces **1143** and **1152** of the insert **1014** are received within the cylindrical surfaces **1086** and **1090** of the receiver. With particular reference to FIGS. **74** and **75**, the receiver thin walls of the sloping surface **1077** located about the arched through bore portion

1075 are then crimped inwardly toward the axis B by inserting a tool (not shown) into the receiver apertures **1074**, the tool pressing the sloped surface walls **1077** until the inner wall surfaces **1087** engage the insert **1014** at the grooves **1160** and **1161** formed into the outer cylindrical surface **1143** of each of the insert arms **1137**. The crimping of the opposed wall surfaces **1087** into the groves **1160** and **1161** keeps the insert **1014** U-shaped channel **1141** substantially aligned with the receiver U-shaped channel **1064**, but allows for upward movement of the insert **1014** along the receiver axis B during bottom loading of the shank **1004** as shown in FIG. **77**, for example. Thus, the crimping of the receiver walls **1077** prohibits rotation of the insert **1104** about the receiver axis B but allows for limited axial movement of the insert **1014** with respect to the receiver **1010** along the axis B when some force is exerted to slide the crimped surface **1087** up or down along the grooves **1160** and **1161**. The insert **1014** arms **1137** are fully captured within the receiver **1010** by the guide and advancement structure **1072** prohibiting movement of the insert **1014** up and out through the receiver opening **1066** as well as by the retainer **1012** and the receiver annular surface **1104** located in the receiver **1010** base **1060** below the insert **1014**.
(177) In some embodiments of the invention, top or side surfaces of the insert **1014** may include a resilient projection or projections for temporarily frictionally engaging with an inner surface of the receiver **1010** to hold the insert **1014** in an upper portion of the receiver **1010** during some of the assembly steps, also providing a frictional but slidable fit between the insert **1014** and the receiver **1010**. In the illustrated embodiment, the insert **1014** is substantially freely slidable in the upper portion of the receiver **1010** in an axial direction, and sized an shaped so that the insert **1014** is located above the cylindrical surface **1099** during expansion of the retainer **1012** about the shank head **1008**, the surface **1099** functioning as an expansion chamber or recess for the retainer **1012**.
(178) At this time, the receiver, insert and retainer combination are ready for shipping to an end user, with both the compression insert **1014** and the retainer **1012** captured within the receiver **1010** in a manner that substantially prevents movement or loss of such parts out of the receiver **1010**. The receiver **1010**, compression insert **1014** and the retainer **1012** combination may now be assembled with the shank **1004** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **1004** as shown, for example, in FIG. **76**, with the shank axis A and the receiver axis B either being aligned during assembly as shown in FIG. **77** and most of the drawings figures illustrating the assembly process, or the axes being at an angle with respect to one another as shown in FIG. **76**.
(179) As illustrated in FIG. **76**, the bone screw shank **1004** or an entire assembly **1001** made up of the assembled shank **1004**, receiver **1010**, retainer **1012** and compression insert **1014**, is screwed into a bone, such as the vertebra **1017**, by rotation of the shank **1004** using a suitable driving tool (not shown) that operably drives and rotates the shank body **1006** by engagement thereof at the internal drive **10046**. Specifically, the vertebra **1017** may be pre-drilled to minimize stressing the bone and have a guide wire (not shown) inserted therein to provide a guide for the placement and angle of the shank **1004** with respect to the vertebra. A further tap hole may be made using a tap with the guide wire as a guide. Then, the bone screw shank **1004** or the entire assembly **1001** is threaded onto the guide wire utilizing the cannulation bore **1050** by first threading the wire into the opening at the bottom **1028** and then out of the top opening at the drive feature **1046**. The shank **1004** is then driven into the vertebra using the wire as a placement guide. It is foreseen that the shank and other bone screw assembly parts, the rod **1021** (also having a central lumen in some embodiments) and the closure top **1018** (also with a central bore) can be inserted in a percutaneous or minimally invasive surgical manner, utilizing guide wires. When the shank **1004** is driven into the vertebra **1017** without the remainder of the assembly **1001**, the shank **1004** may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer.
(180) With reference to FIGS. **76** and **77**, the pre-assembled receiver, insert and retainer are placed above the shank upper portion **1008** until the shank upper portion is received within the opening

1110. With particular reference to FIGS. **77** and **78**, as the shank upper portion **1008** is moved into the interior **1061** of the receiver base, the shank upper portion **1008** presses the retainer **1012** upwardly into the recess partially defined by the cylindrical surface **1099** and partially by the annular surface **1095**. With particular reference to FIGS. **78** and **79**, as the portion **1008** continues to move upwardly toward the channel **1064**, the top surface **1122** of the retainer **1012** abuts against the receiver annular surface **1095**, stopping upward movement of the retainer **1012** and forcing outward movement of the retainer **1012** towards the cylindrical surface **1099** defining the receiver expansion recess as the spherical surface **1034** continues in an upward direction. With reference to FIGS. **80** and **81**, the retainer **1012** begins to contract about the spherical surface **1034** as the center of the sphere of the head **1008** passes beyond the center of the retainer expansion recess defined by the surface **1099**. At this time also, the spherical surface **1034** moves into engagement with the insert **1014** collet panels **1155** at the panel inner planar surfaces **1173**, the panels **1155** initially expanding slightly outwardly to receive the surface **1034**. The panels **1155** press outwardly toward the surface **1090** that provides enough clearance for the spherical surface **1034** to enter into full frictional engagement with the panel inner surfaces **1173** as shown in FIG. **81**. At this time, the insert **1014** and the surface **1034** are in a fairly tight friction fit, the surface **1034** being pivotable with respect to the insert **1014** with some force. Thus, a tight, non-floppy ball and socket joint is now created between the insert **1014** and the shank upper portion **1008**.

(181) The shank **1004** and attached insert **1014** may then be manipulated further downwardly into a desired position for receiving the rod **1021** or other longitudinal connecting member by either an upward pull on the receiver **1010** or, in some cases, by driving the shank **1004** further into the vertebra **1017**. Also, in some embodiments, when the receiver **1010** is pre-assembled with the shank **1004**, the entire assembly **1001** may be implanted at this time by inserting the driving tool (not shown) into the receiver and the shank drive **1046** and rotating and driving the shank **1004** into a desired location of the vertebra **1017**. Also, at this time, the receiver **1010** and may be articulated to a desired angular position with respect to the shank **1004**, such as that shown in FIG. **83**, but prior to insertion of the rod or closure top, that will be held, but not locked, by the frictional engagement between the retainer **1012** and the shank upper portion **1008**.

(182) With reference to FIGS. **82** and **83**, the rod **1021** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **1001**. The closure structure **1018** is then inserted into and advanced between the arms **1062** of each of the receivers **1010**. The closure structure **1018** is rotated, using a tool engaged with the inner drive **1186** until a selected pressure is reached at which point the rod **1021** engages the U-shaped seating surface **1142** of the compression insert **1014**, pressing the insert stepped shank gripping surfaces **1170** against the shank spherical surface **1034**, the edges of the stepped surfaces penetrating into the spherical surface **1034** and also pressing the shank upper portion **1008** into locked frictional engagement with the retainer **1012**. Specifically, as the closure structure **1018** rotates and moves downwardly into the respective receiver **1010**, the rim **1190** engages and penetrates the rod surface **1022**, the closure structure **1018** pressing downwardly against and biasing the rod **1021** into compressive engagement with the insert **1014** that urges the shank upper portion **1008** toward the retainer **1012** and into locking engagement therewith, the retainer **1012** frictionally abutting the surface **1104** and expanding outwardly against the cylindrical surface **1101**. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **1006** with respect to the receiver **1010**.

(183) With reference to FIGS. **84-89**, an alternative lock-and-release compression insert **1014′** is illustrated for use with the shank **1004**, receiver **1010**, retainer **1012**, closure top **1018** and rod **1021** previously described herein, the resulting assembly identified as an assembly **1001′** in FIGS. **89** and **90**, for example. The insert **1014′** is identical or substantially similar to the insert **1014** previously described herein, with the exception that the insert **1014′** is sized for a locking, frictional interference fit with the receiver **1010**; specifically, a locking interference between the cylindrical

inner surface **1090** of the receiver **1010** and a lower portion of the outer cylindrical surfaces **1143′** of the insert arms **1137′** as will be described in greater detail below. The illustrated insert **1014′** also differs from the insert **1014** in that the insert **1014′** includes a pair of opposed through bores **1159′** for receiving tooling as will be described in greater detail below.
(184) Thus, with reference to FIGS. **84-86**, the locking insert **1014** includes a body **1136′**, a pair of opposed arms **1137′**, a pair of crown collet extensions **1138′**, a through bore **1140′**, a u-shaped saddle surface **1141′** with a seat **1142′**, arm outer surfaces **1143′**, arm top surfaces **1144′**, collet extension bottom surfaces **1148′**, arms rim or ledge surfaces **1150′**, annular lower body surfaces **1151′**, lower outer cylindrical surfaces **1152′**, slots **1154′**, panels **1155′** formed by the slots, notches **1156′** having an upper sloping surface **1157′** and a lower planar surface **1158′**, a pair of grooves **1160′** and **1161′**, having respective bottom surfaces **1162′** and **1163′**, inner planar opposed surfaces **1165′**, inner cylindrical surfaces **1166′**, an inner spherical profile **1168′** with a gripping surface portion **1170′**, curved inner surfaces **1172′** and collet planar inner gripping friction fit surfaces **1173′** that are the same or substantially similar in form and function to the respective body **1136**, pair of opposed arms **1137**, pair of crown collet extensions **1138**, through bore **1140**, u-shaped saddle surface **1141** with seat **1142**, arm outer surfaces **1143**, arm top surfaces **1144**, collet extension bottom surfaces **1148**, arms rim or ledge surfaces **1150**, annular lower body surfaces **1151**, lower outer cylindrical surfaces **1152**, slots **1154**, panels **1155** formed by the slots, notches **1156** having upper sloping surfaces **1157** and a lower planar surface **1158**, pair of grooves **1160** and **1161** having respective bottom surfaces **1162** and **1163**, inner planar opposed surfaces **1615**, inner cylindrical surfaces **1166**, inner spherical profile **1168** with a gripping surface portion **1170**, curved inner surfaces **1172** and planar friction fit gripping surfaces **1173** previously described herein with respect to the insert **1014**.
(185) The insert **1014′** outer cylindrical surfaces **1143′** that are located below the tooling notches **1156′** and at or near the ledge or rim **1150′** are sized and shaped for a locking interference fit with the receiver **1010** at the cylindrical surface **1090** that partially defines the inner cavity **1061**. In other words, a width or diameter measured between arm surfaces **1143′** at or directly above the rim **1150′** is sized large enough to require that the insert **1014′** must be forced into the space defined by the cylindrical surface **1090** starting at the edge defined by the receiver surface **1088** and the surface **1090** by a tool or tools or by the closure top **1018** forcing the rod **1021** downwardly against the insert **1014′** with sufficient force to interferingly lock the insert **1014′** into the receiver **1010** at the cylindrical surface **1090**.
(186) With reference to FIGS. **87-90**, the insert **1014′** is assembled with the receiver **1010**, retainer **1012**, shank **1004**, rod **1021** and closure top **1018**, in a manner the same as previously described above with respect to the assembly **1001**, resulting in an assembly **1001′**, with the exception that the insert **1014′** must be forced downwardly into a locking interference fit with the receiver **1010** when the shank **1004** is locked in place, as compared to the easily sliding relationship between the insert **1014** and the receiver **1010**. In particular, prior to assembly with the rod **1021** and the closure top **1018**, the compression insert **1014′** outer cylindrical surfaces **1152′** are slidingly received by receiver cylindrical surface **1090**, but the surfaces **1143′** are not. The insert **1014′** is thus prohibited from moving any further downwardly at the receiver surface **1088** unless forced downwardly by a locking tool or by the closure top pressing downwardly on the rod that in turn presses downwardly on the insert **1014′** as shown in FIGS. **89** and **90**. With further reference to FIG. **89**, at this time, the receiver **1010** may be articulated to a desired angular position with respect to the shank **1004**, such as that shown in FIG. **96**, for example, that will be held, but not locked, by the frictional engagement between the retainer **1012** collet panels and the shank upper portion **1008**.
(187) The rod **1021** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **1001′**. The closure structure **1018** is then inserted into and advanced between the arms **1062** of each of the receivers **1010**. The closure structure **1018** is rotated, using a tool engaged with the inner drive **1186** until a selected pressure is reached at which

point the rod **1021** engages the U-shaped seating surface **1142′** of the compression insert **1014′**, pressing the insert stepped shank gripping surfaces **1170′** against and into the shank spherical surface **1034**, the edges of the stepped surfaces **1170′** penetrating into the spherical surface **1034**, pressing the shank upper portion **1008** into locked frictional engagement with the retainer **1012**. Specifically, as the closure structure **18** rotates and moves downwardly into the respective receiver **1010**, the rim **1190** engages and penetrates the rod surface **1022**, the closure structure **1018** pressing downwardly against and biasing the rod **1021** into compressive engagement with the insert **1014′** that urges the shank upper portion **1008** toward the retainer **1012** and into locking engagement therewith, the retainer **1012** frictionally abutting the surface **1104** and expanding outwardly against the cylindrical surface **1101**. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **1006** with respect to the receiver **1010**. Tightening the helical flange form to approximately 100 inch pounds can create around 900 to 1000 pounds of force and it has been found that the interference fit created between the cylindrical surfaces **1143′** of the insert **1014′** and the cylindrical surface **1090** of the receiver can be overcome at between about 500 to 700 inch pounds depending on manufacturing tolerance issues between the parts. So, as the closure structure **1018** and the rod **1021** press the insert **1014′** downwardly toward the base of the receiver **1010**, the insert surfaces **1143′** are forced into the receiver at the edge defined by the receiver annular surface **1088** and the cylindrical surface **1090**, thus forcing and fixing the insert **1014** into frictional interference engagement with the receiver at and along the surface **1090**, while not substantially affecting the locking of the polyaxial mechanism itself.

(188) With reference to FIG. **91**, at this time, the closure top **1018** may be loosened or removed and/or the rod **1021** may be adjusted and/or removed and the frictional engagement between the insert **1014′** and the receiver **1010** at the insert surfaces **1143′** will remain locked in place, advantageously maintaining a locked angular position of the shank **1004** with respect to the receiver **1010**.

(189) With further reference to FIGS. **91** and **92**, at this time, another rod, such as the deformable rod **1221** and cooperating alternative closure top **1218** may be loaded onto the already locked-up assembly to result in an alternative assembly **1201′**. As mentioned above, the closure drive **1286** may advantageously be made smaller than the drive of the closure **1018**, such that the deformable rod **1221** is not unduly pressed or deformed during assembly since the polyaxial mechanism is already locked.

(190) With reference to FIGS. **93-95**, a two-piece tool, generally **1600**, is illustrated for releasing the insert **1014′** from the receiver **1010**. The tool **1600** includes an inner flexible tube-like structure with opposed inwardly facing prongs **1612** located on either side of a through-channel **1616**. The channel **1616** may terminate at a location spaced from the prongs **1612** or may extend further upwardly through the tool, resulting in a two-piece tool **1610**. The tool **1600** includes an outer, more rigid tubular member **1620** having a smaller through channel **1622**. The member **1620** slidingly fits over the tube **1610** after the flexible member **1610** prongs **1612** are flexed outwardly and then fitted over the receiver **1010** and then within through bores of the opposed apertures **1074** of the receiver **1010** and aligned opposed bores **1159′** located on arms of the insert **1014′**. In FIG. **93**, the tool **1600** is shown during the process of unlocking the insert **1014′** from the receiver **1010** with the outer member **1620** surrounding the inner member **1610** and holding the prongs **1612** within the receiver **1010** and insert **1014′** apertures while the tool **1600** is pulled upwardly away from the shank **1004**. It is foreseen that the tool **1600** may further include structure for pressing down upon the receiver **1010** while the prongs and tubular member are pulled upwardly, such structure may be located within the tool **1600** and press down upon the top surfaces **1073** of the receiver arms, for example.

(191) Alternatively, another manipulation tool (not shown) may be used that is inserted into the receiver at the opening **1066** and into the insert channel formed by the saddle **1141′**, with prongs or extensions thereof extending outwardly into the insert through bores **1159′**; a piston-like portion of

the tool thereafter pushing directly on the shank upper portion **1008**, thereby pulling the insert **1014′** away from the receiver surface **1090** and thus releasing the polyaxial mechanism. At such time, the shank **1004** may be articulated with respect to the receiver **1010**, and the desired friction fit returns between the retainer **1012** and the shank surface **1034**, so that an adjustable, but non-floppy relationship still exists between the shank **1004** and the receiver **1010**. If further disassembly if the assembly is desired, such is accomplished in reverse order to the procedure described previously herein for the assembly **1001**.
(192) With reference to FIGS. **96-98**, another manipulation tool, generally **1700**, is illustrated for independently locking the insert **1014′**, or, in some embodiments, temporarily locking the non-locking insert **1014**, to the receiver **1010**. The tool **1700** includes a pair of opposed arms **1712**, each having an engagement extension **1716** positioned at an angle with respect to the respective arm **1712** such that when the tool is moved downwardly toward the receiver, one or more inner surfaces **1718** of the engagement extension **1716** slide along the surfaces **1077** of the receiver and along **1157′** of the insert **1014′** to engage the insert **1014′**, with a surface **1720** pressing downwardly on the insert surfaces **1158′**, pushing the cylindrical arm surfaces **1143′** into an interference locking fit within the receiver cylindrical surface **1090**. As shown in FIG. **98**, when the insert **1014′** is locked against the receiver **1010**, the tool bottom surfaces **1720** do not bottom out on the receiver surfaces **7105′**, but remained spaced therefrom. In the illustrated embodiment, the surface **1718** is slightly rounded and each arm extension **1716** further includes a planar lower surface **1722** that creates an edge with the bottom surface **1720** for insertion and gripping of the insert **1014′** at the juncture of the surface **1157′** and the surface **1158′**. The tool **1700** may include a variety of holding and pushing/pulling mechanisms, such as a pistol grip tool, that may include a ratchet feature, a hinged tool, or, a rotatably threaded device, for example.
(193) It is to be understood that while certain forms of the present invention have been illustrated and described herein, it is not to be limited to the specific forms or arrangement of parts described and shown.

Claims

1. A pivotal bone anchor assembly for securing an elongate rod to a bone of a patient with a closure top via independent provisional locking with an insert compressing tool, the pivotal bone anchor assembly comprising: a receiver housing having a vertical centerline axis, a base portion defining a cavity communicating with a bottom of the receiver housing through a bottom opening, and an upper portion defining a channel having a transverse axis perpendicular to the vertical centerline axis and configured to receive the elongate rod, the channel communicating with the cavity and the upper portion including inner surfaces with a closure top mating structure formed therein; a shank having longitudinal axis, a head portion with a partial spherical upper surface, and an anchor portion opposite the head portion configured for attachment to the bone, the head portion configured for positioning within the cavity of the receiver housing with the shank being pivotal with respect to the receiver housing in an unlocked configuration; and an insert at least partially positionable into the channel of the receiver housing, the insert having a contoured seating surface configured to receive and engage the elongate rod, the contoured seating surface having a length alignable with the transverse axis of the channel of the receiver housing and a width perpendicular to the length, the length of the contoured seating surface being greater than the width such that the insert can only be positioned in the receiver housing in one predetermined orientation or the exact reverse thereof, the insert further including at least one upward-facing surface located laterally outward from the contoured seating surface, wherein the at least one upward-facing surface of the insert is configured for releasable downward compressive engagement by the insert compressing tool after the elongate rod is received within the channel of the receiver housing and prior to a final locking of the assembly with the closure top, so as to independently provisionally lock the shank

from pivoting with respect to the receiver housing.

2. The pivotal bone anchor assembly of claim 1, wherein the at least one upward-facing surface of the insert further comprises a pair of upward-facing surfaces positioned laterally opposite of each other.

3. The pivotal bone anchor assembly of claim 1, wherein the insert further includes a lower portion configured to directly engage the partial spherical upper surface of the head portion of the shank.

4. The pivotal bone anchor assembly of claim 1, wherein the insert is configured for positioning within the receiver housing prior to the shank.

5. The pivotal bone anchor assembly of claim 1, wherein the shank is cannulated with a central bore extending an entire length of the shank along the longitudinal axis.

6. The pivotal bone anchor assembly of claim 1 and further comprising the elongate rod and the closure top, wherein the closure top includes a continuous guide and advancement structure configured to threadably engage the closure top mating structure of the upper portion of the receiver housing to advance the closure top into the channel of the receiver housing upon rotation of the closure top relative to the receiver housing.

7. The pivotal bone anchor assembly of claim 6, wherein the closure top is configured to compress a top surface of the elongate rod when the elongate rod is received within the channel of the receiver housing to provide the final locking of the assembly.

8. A pivotal bone anchor system comprising the pivotal bone anchor assembly of claim 6 and the insert compressing tool, wherein the insert compressing tool is configured to non-threadably and slidably engage the upper portion of the receiver housing while applying the releasable downward compressive engagement to the at least one upward-facing surface of the insert so as to provide the independent provisional locking of the shank with respect to the receiver housing.

9. The pivotal bone anchor system of claim 8, wherein the insert compressing tool further comprises downward-facing surfaces complementary with the at least one upward-facing surface of the insert.

10. The pivotal bone anchor system of claim 8, wherein the at least one upward-facing surface of the insert is configured to be released by the insert compressing tool after the assembly is locked by the elongate rod and the closure top in a final locked configuration.

11. A pivotal bone anchor assembly for securing an elongate rod to a bone of a patient with a closure top via independent provisional locking with an insert compressing tool, the pivotal bone anchor assembly comprising: a receiver housing comprising a vertical centerline axis, a base portion defining a cavity communicating with a bottom of the receiver housing through a bottom opening, and an upper portion defining a channel having a transverse axis perpendicular to the vertical centerline axis and configured to receive the elongate rod, the channel communicating with the cavity and the upper portion including inner surfaces with a closure top mating structure formed therein; a shank comprising a longitudinal axis, an at least partially spherical shank head at a proximal end, and an anchor portion opposite the shank head configured for attachment to the bone, the shank head being positionable within the cavity of the receiver housing with the shank being pivotal with respect to the receiver housing in an unlocked configuration; and an insert at least partially positionable into the channel of the receiver housing, the insert comprising an upward facing saddle surface configured to receive and engage the elongate rod and a central drive opening configured for receiving a drive tool therethrough, the saddle surface having a length alignable with the transverse axis of the channel of the receiver housing and a width perpendicular to the length, the length of the saddle surface being greater than the width such that the insert can only be positioned in the receiver housing in one predetermined orientation or the exact reverse thereof, the insert further including opposite upward-facing surfaces located laterally outward from the upward facing saddle surface on each side thereof, wherein at least one of the opposite upward-facing surfaces of the insert is configured for releasable downward compressive engagement by the insert compressing tool after the elongate rod is received within the channel of the receiver housing

and prior to a final locking of the assembly with the closure top, so as to independently provisionally lock the shank from pivoting with respect to the receiver housing prior to a locking of the shank with respect to the receiver housing with the elongate rod and closure top.

12. The pivotal bone anchor assembly of claim 11, wherein the insert further includes a lower portion configured to directly engage the at least partially spherical shank head.

13. The pivotal bone anchor assembly of claim 11, wherein the insert is configured for positioning within the receiver housing prior to the shank.

14. The pivotal bone anchor assembly of claim 11, wherein the shank is cannulated with a central bore extending an entire length of the shank along the longitudinal axis.

15. The pivotal bone anchor assembly of claim 11 and further comprising the elongate rod and the closure top, wherein the closure top includes a continuous guide and advancement structure configured to threadably engage the closure top mating structure of the upper portion of the receiver housing to advance the closure top into the channel of the receiver housing upon rotation of the closure top relative to the receiver housing.

16. The pivotal bone anchor assembly of claim 15, wherein the closure top is configured to compress a top surface of the elongate rod when the elongate rod is received within the channel of the receiver housing to provide the final locking of the assembly.

17. A pivotal bone anchor system comprising the pivotal bone anchor assembly of claim 15 and the insert compressing tool, wherein the insert compressing tool is configured to non-threadably and slidably engage the upper portion of the receiver housing while applying the releasable downward compressive engagement to the at least one upward-facing surface of the insert so as to provide the independent provisional locking of the shank with respect to the receiver housing.

18. The pivotal bone anchor system of claim 17, wherein the insert compressing tool further comprises downward-facing surfaces complementary with the at least one upward-facing surface of the insert.

19. The pivotal bone anchor system of claim 17, wherein the at least one upward-facing surface of the insert is configured to be released by the insert compressing tool after the assembly is locked by the elongate rod and the closure top in a final locked configuration.
